MAX PLANCK INSTITUTE
FOR INTELLIGENT SYSTEMS

# PSYCHOMETRICS FOR LARGE LANGUAGE MODEL EVALUATION

## LESSONS AND CHALLENGES

**New templates**



## Education

- B.Sc. Business Informatics (TU Berlin)
  - Research Intern: MPI-SWS
  - Student Researcher: TU Berlin
- M.Sc. Information Systems Management (TU Berlin)
  - Student Research Fellow: Harvard Business School
  - Student Researcher: Zalando Research
- Doctoral Candidate at MPI-IS (IMPRS-IS)

## Past & Ongoing Research

Universitat Pompeu Fabra Barcelona

Fair Ranking

MAX PLANCK INSTITUTE FOR SOFTWARE SYSTEMS

Fair Matching

HARVARD BUSINESS SCHOOL

Human-AI Collaboration

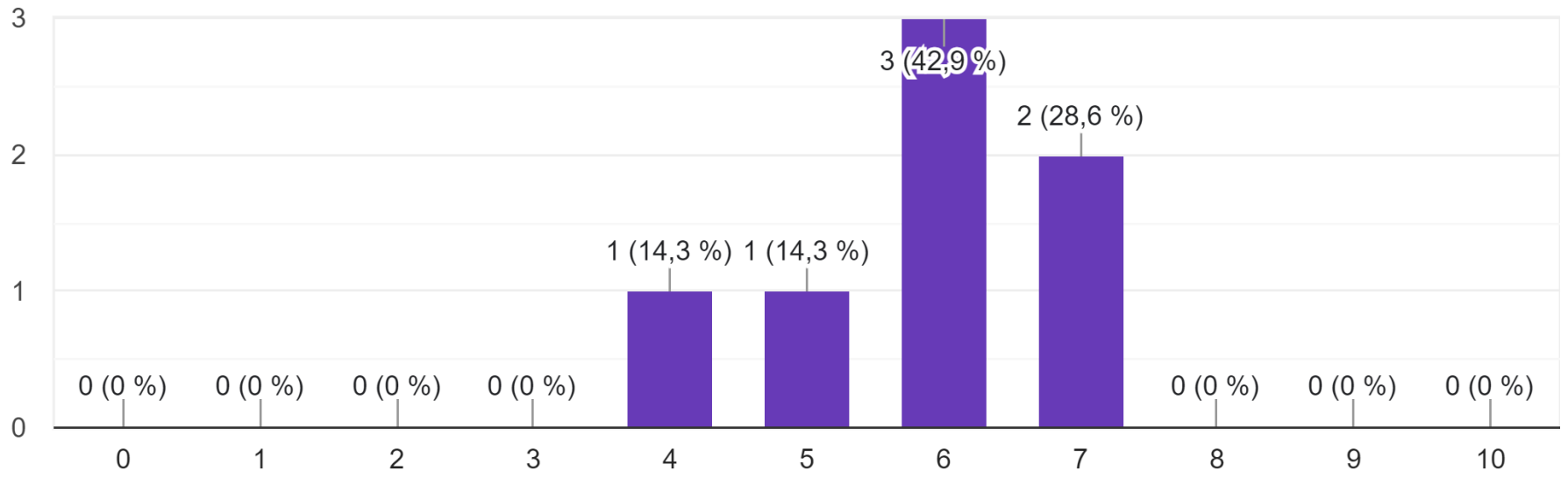
EBERHARD KARLS UNIVERSITÄT TÜBINGEN

Psychometrics for ML

# SURVEY RESULTS

On a scale of 0 to 10, how well do you know Python?

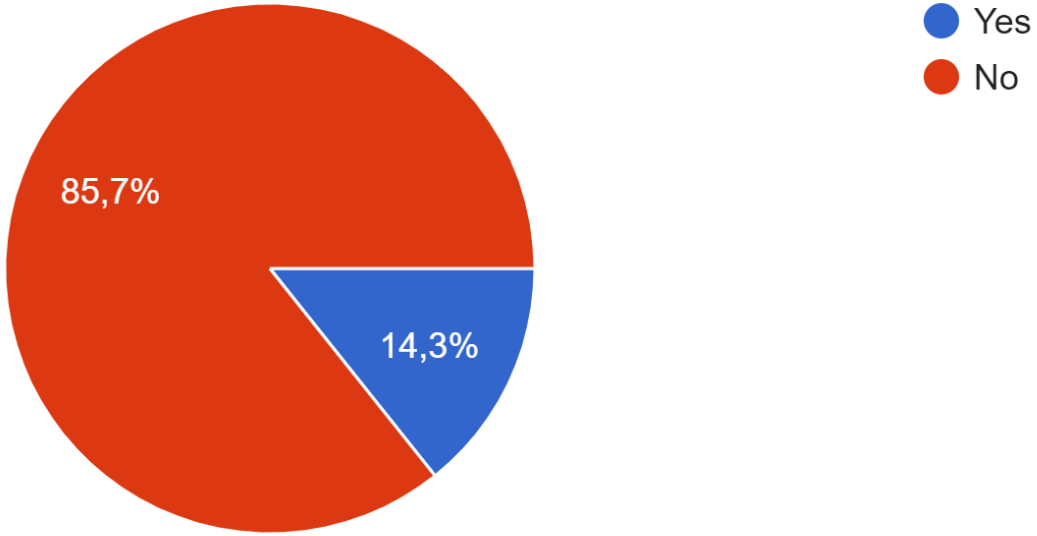
7 Antworten



# SURVEY RESULTS

Have you used PyTorch?

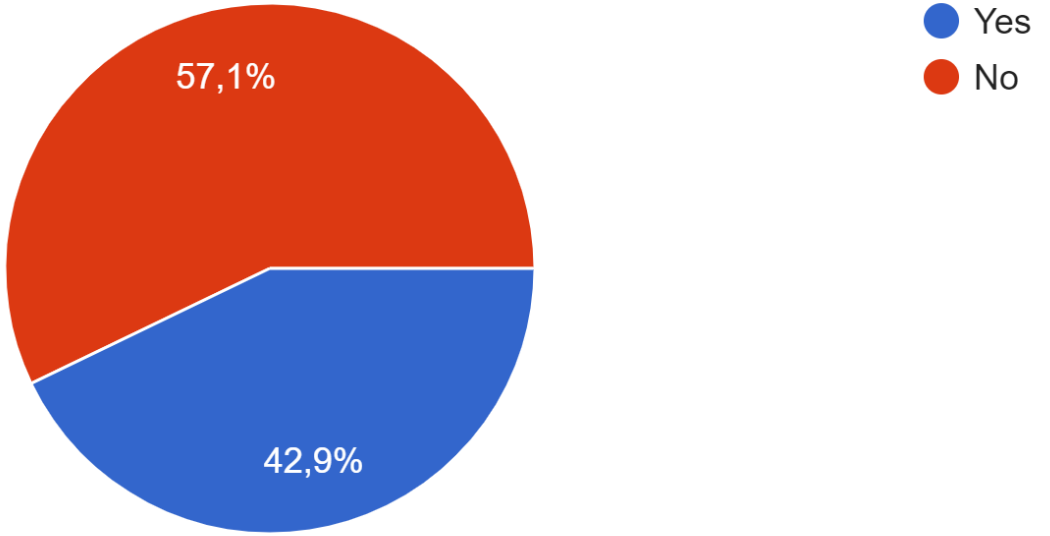
7 Antworten



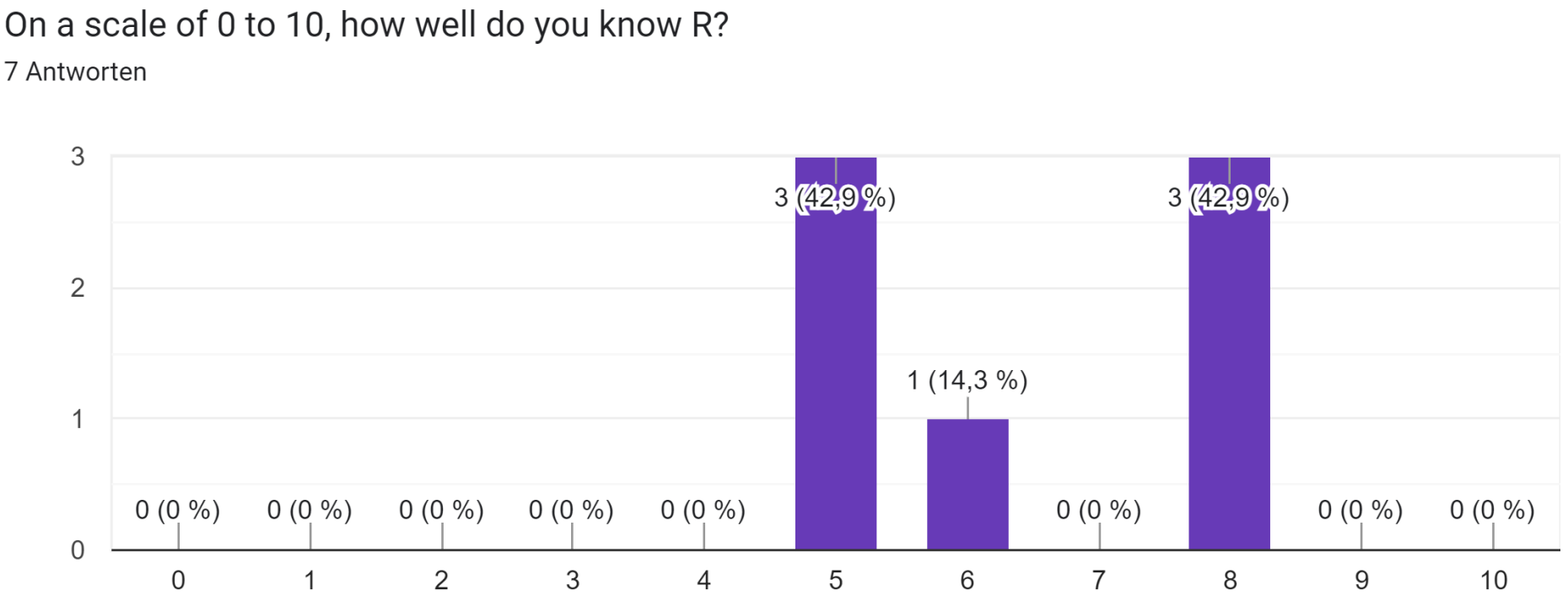
# SURVEY RESULTS

Have you used TensorFlow, Keras or other ML libraries other than PyTorch?

7 Antworten



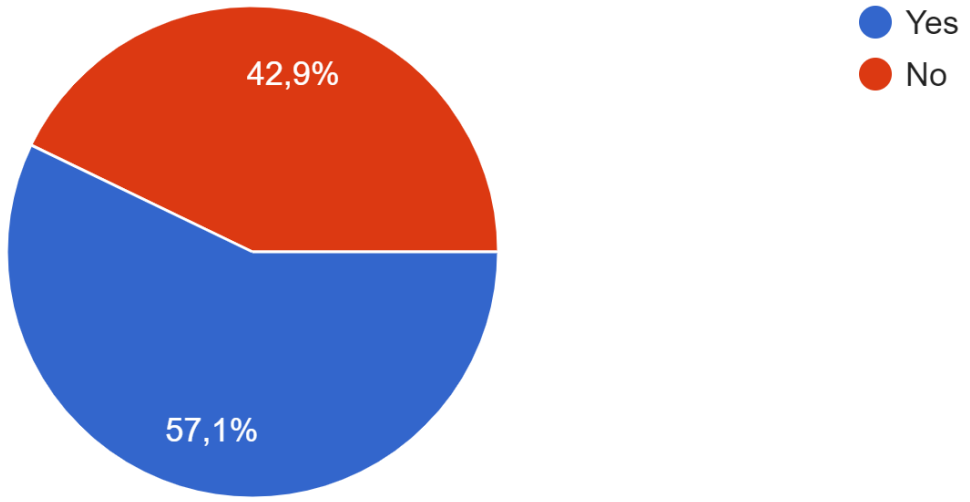
# SURVEY RESULTS



# SURVEY RESULTS

Have you prompted an LLM without a graphical UI? (OpenAI API, transformers library, or comparable)

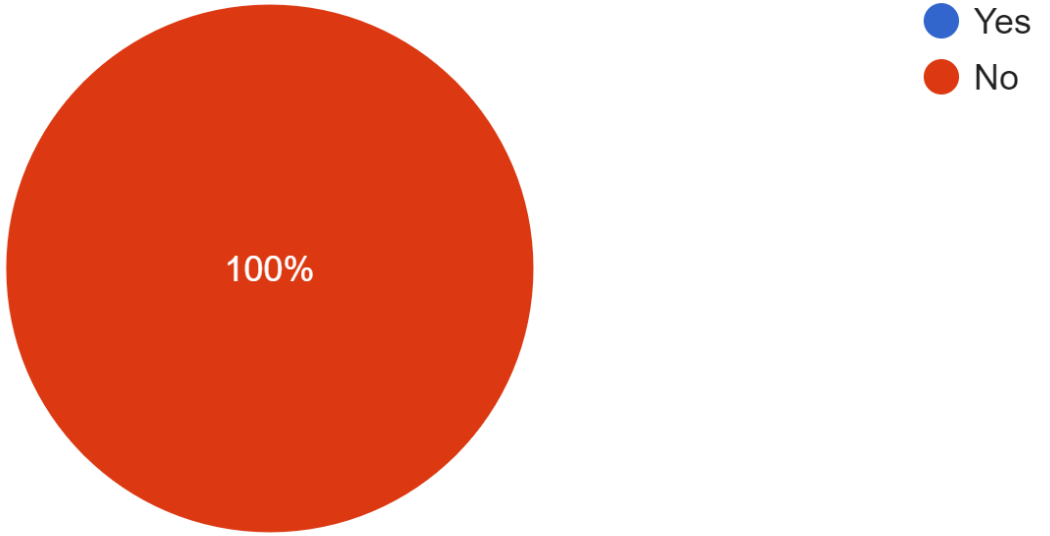
7 Antworten



# SURVEY RESULTS

Have you fine-tuned an LLM (not prompt engineering but actual weight updates or adapters)?

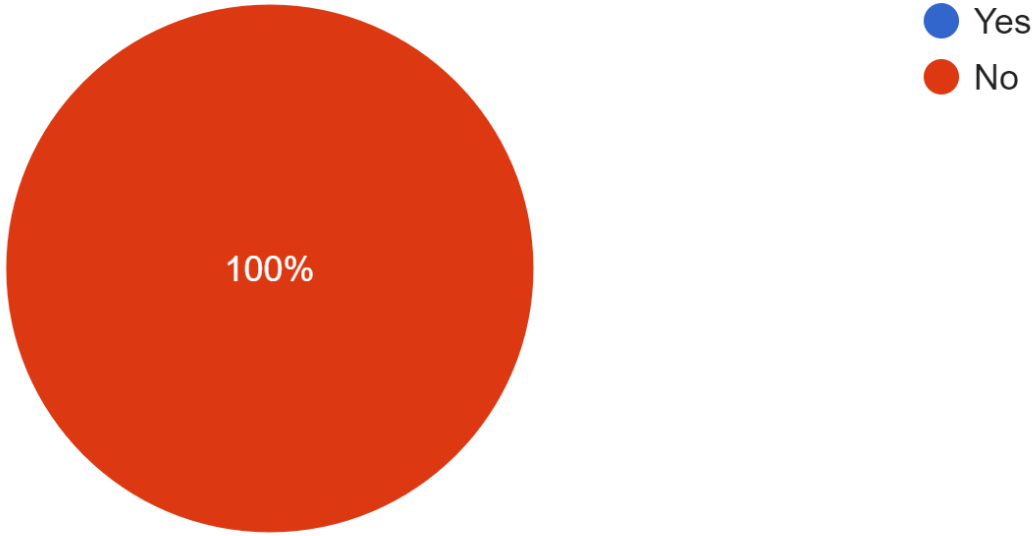
7 Antworten



# SURVEY RESULTS

Do you have experience with axolotl or deepspeed?

7 Antworten



# RULES FOR TODAY

**LLMs and especially Psychometrics + LLMs is cutting edge ML research and probably one of the most novel methods out there. Even machine learning itself is a novel method for most fields. Therefore:**

- **There are now “too basic” questions! (Chances are that I don’t know the answer either and we can find out together)**
- **Please ask for clarification/explanation if I use ML/Computer Science vocabulary.**
- **Please correct me if I am wrong.**

```yaml
base_model: meta-llama/Meta-Llama-3-8B-Instruct
model_type: AutoModelForCausalLM
tokenizer_type: AutoTokenizer

load_in_8bit: false
load_in_4bit: true
strict: false

datasets:
  - path: /fast/tsuehr/sem_data/generated_data_100k.jsonl
    ds_type: json
    type: sharegpt.load_role
psytest: bfi
conditions: personas
model_name: llama3-8b-instruct-qlora
dataset_prepared_path:
val_set_size: 0
output_dir: ./outputs/qlora-out

adapter: qlora
lora_model_dir:

sequence_len: 8192
sample_packing: true
pad_to_sequence_len: true

lora_r: 32
lora_alpha: 16
lora_dropout: 0.05
lora_target_modules:
lora_target_linear: true
lora_fan_in_fan_out:
```

```yaml
wandb_project:
wandb_entity:
wandb_watch:
wandb_name:
wandb_log_model:

gradient_accumulation_steps: 4
micro_batch_size: 2
num_epochs: 3
optimizer: paged_adamw_32bit
lr_scheduler: cosine
learning_rate: 0.0002

train_on_inputs: false
group_by_length: false
bf16: auto
fp16:
tf32: false

gradient_checkpointing: true
early_stopping_patience:
resume_from_checkpoint:
local_rank:
logging_steps: 1
xformers_attention:
flash_attention: true

warmup_steps: 10
evals_per_epoch: 4
eval_table_size:
saves per epoch: 1
```

# AGENDA

1 What is a Large Language Models?

2 How to query an LLM (hands-on)

3 How to train LLMs (without fine-tuning)

4 Ways to fine-tune an LLM

5 How to fine-tune with “smaller” GPUs (optional)

6 Lunch Break

7 A brief history of machine learning benchmarking

8 How to convert LLM answers into benchmarking results

9 Psychometrics for LLM benchmarking – a mistake?

10 IRT for benchmarking optimization

# TECHNICAL SETUP

- Internet access (eduroam?)
- Huggingface account
- Code and data on Github
- Jupyter notebook + other python libs
- R

# AGENDA

## 1 What is a Large Language Models?

# WHAT IS A ~~LARGE~~ LANGUAGE MODEL?

Probability distribution over a sequence of words $x_1, \ldots, x_L$ in a vocabulary $\mathcal{V}$

$$p(x_1, \ldots, x_L)$$

$\mathcal{V}$ = [mouse, cheese, eat, the]

$$p(\text{the, mouse, ate, the, cheese}) = 0.02,$$

$$p(\text{the, cheese, ate, the, mouse}) = 0.01,$$

$$p(\text{mouse, the, the, cheese, ate}) = 0.0001.$$

# WHAT IS A ~~LARGE~~ LANGUAGE MODEL?

## Tokenizers

- Tokenizers convert strings (sequences of characters) to sequences of tokens (i.e., integers)

  “the mouse ate the cheese” → [“the”, “mouse”, “ate”, “the”, “cheese”] → [0, 1, 2, 0, 3]

Two considerations:

- Build the tokenizer: construct a suitable vocabulary
- Apply the tokenizer: tokenize sequence of characters

```
>>> list(tokenizer.get_vocab().keys())[10000:10010]
['Ġpocket', 'ĠInv', 'Download', 'ĠCrime', 'Ġbene', 'ĠGuide', 'ĠAssembly', 'Ġparameters', 'IE', 'ĠAlexander']
```

```
>>> len(tokenizer)
50257
```

```
>>> tokenizer.tokenize("How would GPT2 tokenize this sentence? I don't know!")
['How', 'Ġwould', 'ĠG', 'PT', '2', 'Ġtoken', 'ize', 'Ġthis', 'Ġsentence', '?', 'ĠI',
'Ġdon', "'t", 'Ġknow', '!']
```

```
>>> tokenizer.encode("How would GPT2 tokenize this sentence? I don't know!")
[2437, 561, 402, 11571, 17, 11241, 1096, 428, 6827, 30, 314, 836, 470, 760, 0]
```

Credit: Ricardo Dominguez-Olmedo and Florian E. Dorner

# WHAT IS A ~~LARGE~~ LANGUAGE MODEL?

## Word Embeddings

From word encodings (integers) to word embeddings (vectors)

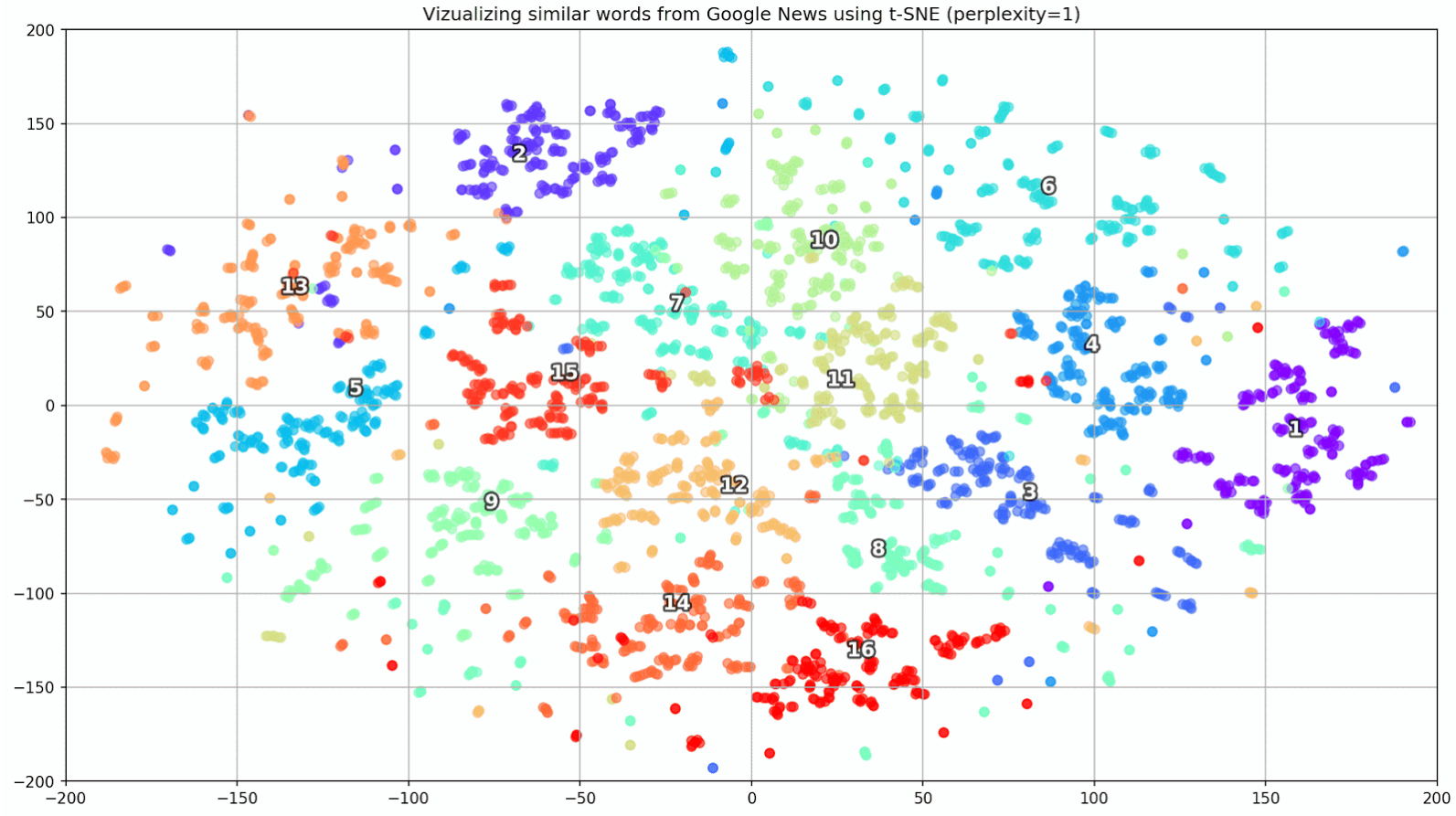
Je suis étudiant → [240, 1243, 32126] →




"Projection" from $\mathbb{R}^{50k} \rightarrow \mathbb{R}^{786}$

Credit: Ricardo Dominguez-Olmedo and Florian E. Dorner

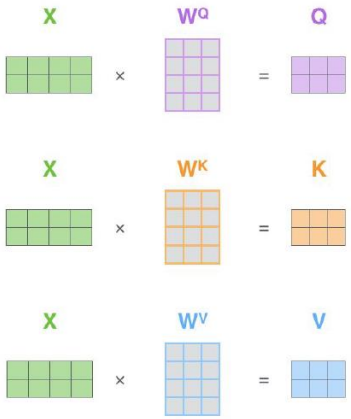
# GOOD EMBEDDINGS, BAD EMBEDDINGS

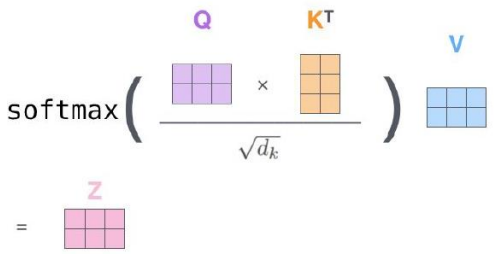


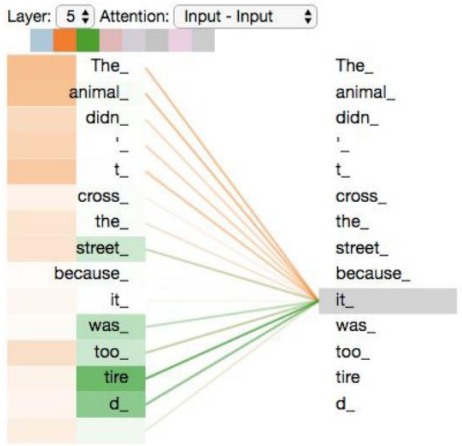
# WHAT IS A ~~LARGE~~ LANGUAGE MODEL?

## Self-Attention

Sequence to sequence processing module

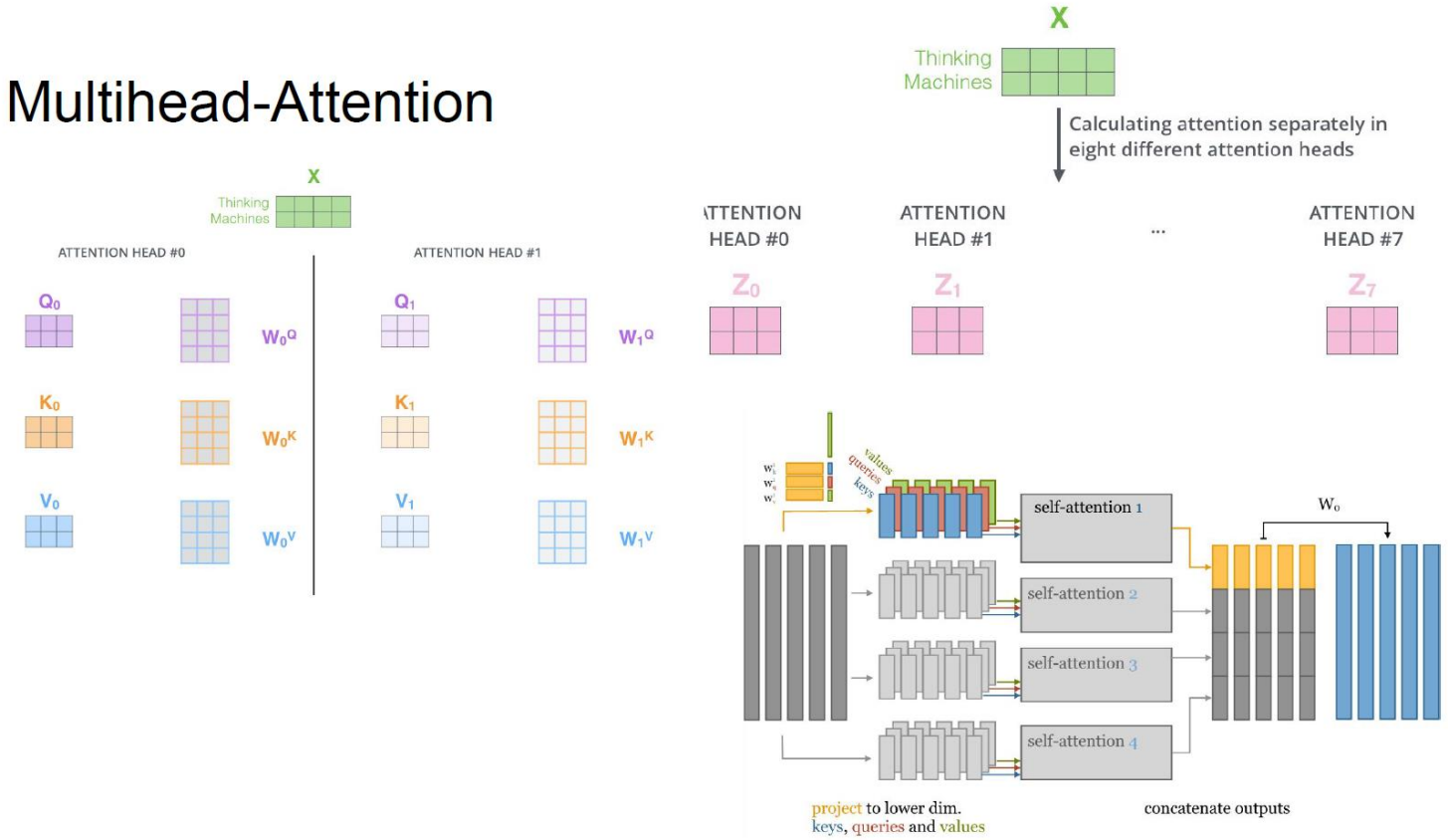


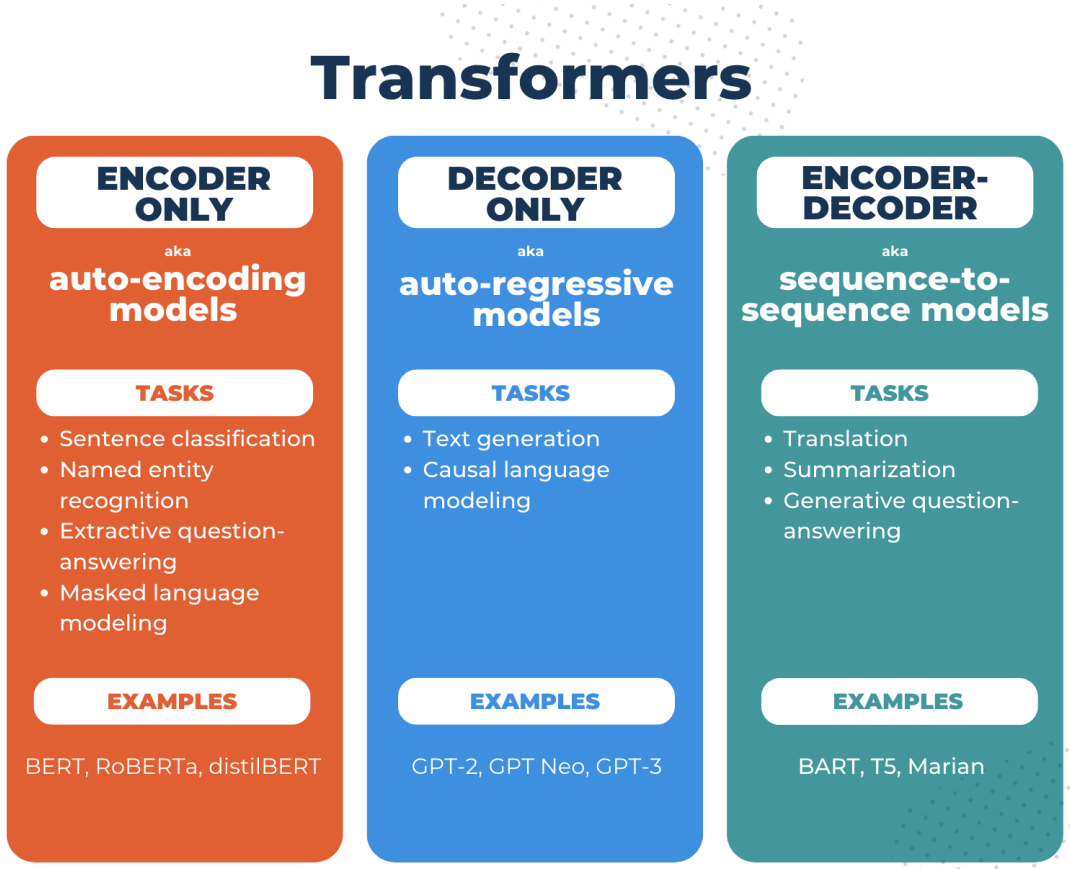

Credit: Ricardo Dominguez-Olmedo and Florian E. Dorner

# WHAT IS A ~~LARGE~~ LANGUAGE MODEL?




Credit: Ricardo Dominguez-Olmedo and Florian E. Dorner

# THE BUILDING BLOCKS OF TRANSFORMERS




Source: https://www.comet.com/site/blog/explainable-ai-for-transformers/

# WHAT IS A ~~LARGE~~ LANGUAGE MODEL?

**ENCODER ONLY**

aka

**auto-encoding models**

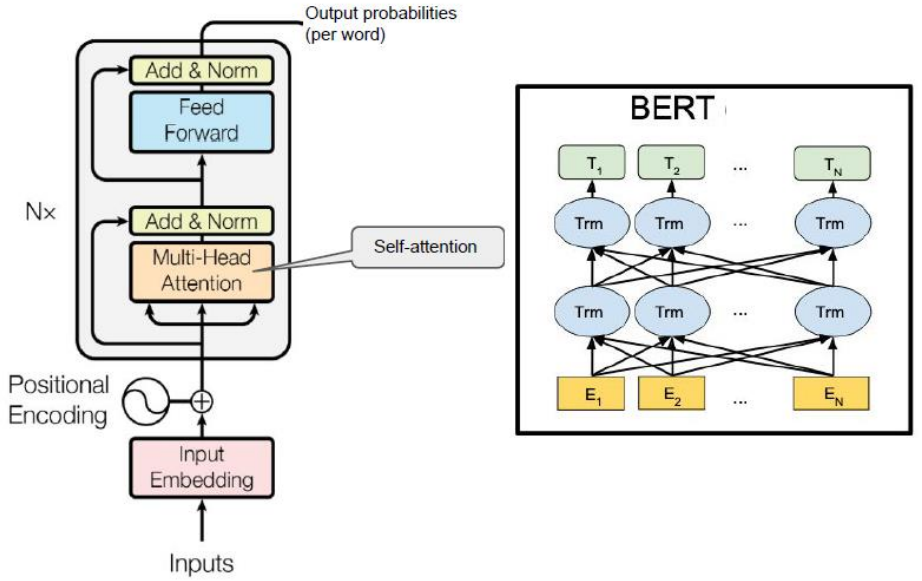
**TASKS**

- Sentence classification
- Named entity recognition
- Extractive question-answering
- Masked language modeling

**EXAMPLES**

BERT, RoBERTa, distilBERT

Encoder only
(BERT, RoBERTa etc.)




$(x_1,...x_n) \rightarrow (\varphi_1(x),...\varphi_n(x))$

Used for: Feature Learning, “Denoising”

# WHAT IS A ~~LARGE~~ LANGUAGE MODEL?

**DECODER ONLY**

aka

**auto-regressive models**

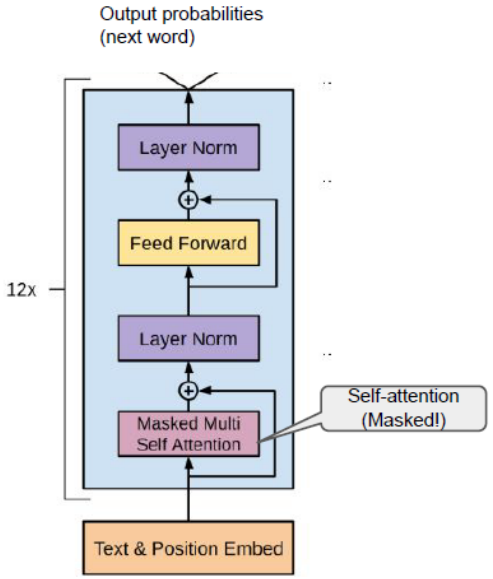
**TASKS**

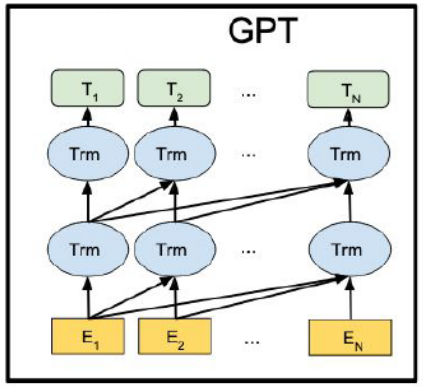
- Text generation
- Causal language modeling

**EXAMPLES**

GPT-2, GPT Neo, GPT-3

Decoder only
(GPT-n)




$(\varphi(x), y_1, \ldots y_k) \rightarrow y_{k+1}$

Text Generation

# WHAT IS A ~~LARGE~~ LANGUAGE MODEL?

**ENCODER-DECODER**

aka

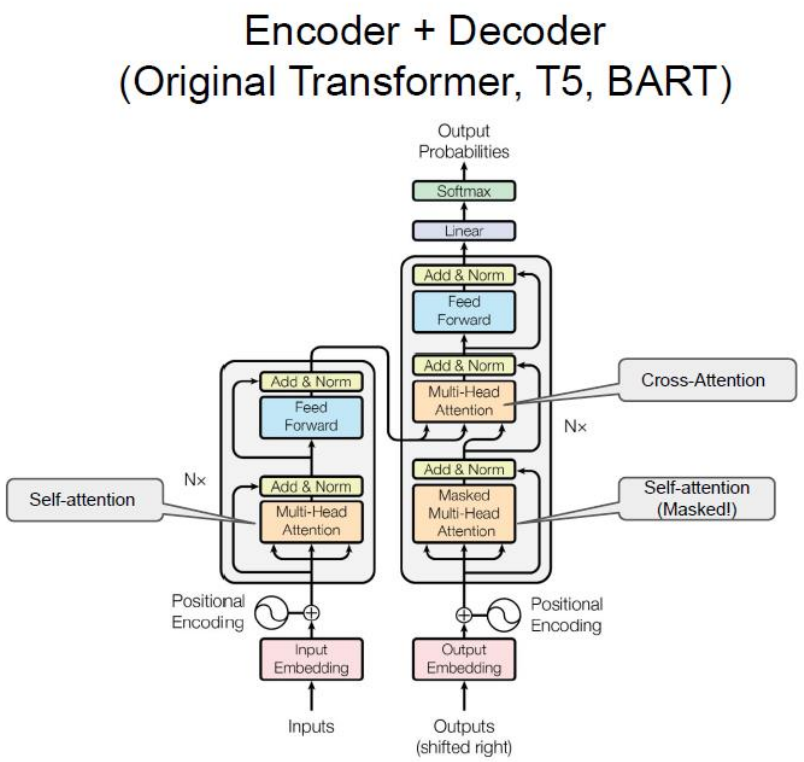
**sequence-to-sequence models**

**TASKS**

- Translation
- Summarization
- Generative question-answering

**EXAMPLES**

BART, T5, Marian



# AGENDA

1 What is a Large Language Models?

2 How to query an LLM (hands-on)

# DEMO: TRAIN/QUERY A FOUNDATION MODEL

- Load pretrained model from huggingface
- Query model
- Show logits
- Show sampling vs argmaxing
- Show temperature

# AGENDA

1 What is a Large Language Models?

2 How to query an LLM (hands-on)

3 How to train LLMs (pre-training)

# HOW TO TRAIN LARGE LANGUAGE MODELS

1. **Supervised learning with SGD**

   **Outcome: “Pre-Trained model” (e.g. GPT-3)**

2. **Supervised Instruction-fine-tuning**
3. **Reward Model Learning**
4. **Reinforcement Learning with human Feedback (RLHF)**

   **Outcome: “Chat-Model” (e.g. ChatGPT)**

Example of ChatGPT

# SUPERVISED LEARNING FOR PRE-TRAINING

**Text in the dataset: “The capital of Spain is Madrid.”**

**Input(X): “The” , Label(Y): “capital”**

**Input: “The capital”, Label: “of”**

**Input: “The capital of”, Label: “Spain”**

**….**

# HOW TO TRAIN LARGE LANGUAGE MODELS

**GPT-n: Causal Language Modelling**

- **Loss :** $-log(p(y_n|y_1, \dots, y_{n-1}))$
- **We minimize the negative log-probability of “Madrid” given “The capital of Spain is”**
- **⇔We maximize the probability of “Madrid” given “The capital of Spain is”**
- **Teacher forcing: always take the ground truth as input for the next prediction, even if we were wrong!**

# HOW TO TRAIN LARGE LANGUAGE MODELS

**Teacher-forcing example:**

**Training example: “Psychometrics is an interesting field of science”**

# HOW TO TRAIN LARGE LANGUAGE MODELS

**Teacher-forcing example:**

**Training example: “Psychometrics is an interesting field of science”**

**Prediction: “Psychometrics … … … … … …”**

# HOW TO TRAIN LARGE LANGUAGE MODELS

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**Training example: “Psychometrics is an interesting field of science”**

**Prediction: “Psychometrics is … … … … …”**

# HOW TO TRAIN LARGE LANGUAGE MODELS

**Teacher-forcing example:**

**Training example: “Psychometrics is an interesting field of science”**

**Prediction: “Psychometrics is the … … … …”**

# HOW TO TRAIN LARGE LANGUAGE MODELS

**Teacher-forcing example:**

**Training example: “Psychometrics is an interesting field of science”**

**Prediction: “Psychometrics is ~~the~~ … … … …”**

# HOW TO TRAIN LARGE LANGUAGE MODELS

**Teacher-forcing example:**

**Training example: “Psychometrics is an interesting field of science”**

**Prediction: “Psychometrics is** an ~~**the**~~ **… … … …”**

# HOW TO TRAIN LARGE LANGUAGE MODELS

**Teacher-forcing example:**

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**Prediction: “Psychometrics is ~~the~~ an interesting … … …”**

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**Teacher-forcing example:**

**Training example: “Psychometrics is an interesting field of science”**

**Prediction: “Psychometrics is ~~the~~** an **interesting discipline … …”**

# HOW TO TRAIN LARGE LANGUAGE MODELS

**Teacher-forcing example:**

**Training example: “Psychometrics is an interesting field of science”**

**Prediction: “Psychometrics is ~~the~~ an interesting ~~discipline~~ … …”**

# HOW TO TRAIN LARGE LANGUAGE MODELS

**Teacher-forcing example:**

**Training example: “Psychometrics is an interesting field of science”**

**Prediction: “Psychometrics is ~~the~~ an interesting ~~discipline~~ field … …”**

# HOW TO TRAIN LARGE LANGUAGE MODELS

**Teacher-forcing example:**

**Training example: “Psychometrics is an interesting field of science”**

**Prediction: “Psychometrics is ~~the~~ an interesting ~~discipline~~ field of …”**

# HOW TO TRAIN LARGE LANGUAGE MODELS

**Teacher-forcing example:**

**Training example: “Psychometrics is an interesting field of science”**

**Prediction: “Psychometrics is ~~the~~ an interesting ~~discipline~~ field of science”**

# HOW TO TRAIN LARGE LANGUAGE MODELS

**A general test-loss for LLMs is Perplexity**

$$Perplexity(x_1, \dots, x_T) = exp(-\frac{1}{T}\sum_{t=1}^{T} log(p(x_t|x_1, \dots, x_{t-1})))$$

- **Perplexity is low if the sentence was very likely**
- **“How perplexed/surprised is the model by the sequence of words”**

# HOW TO TRAIN LARGE LANGUAGE MODELS

- **Generally, more data + larger models = better performance**

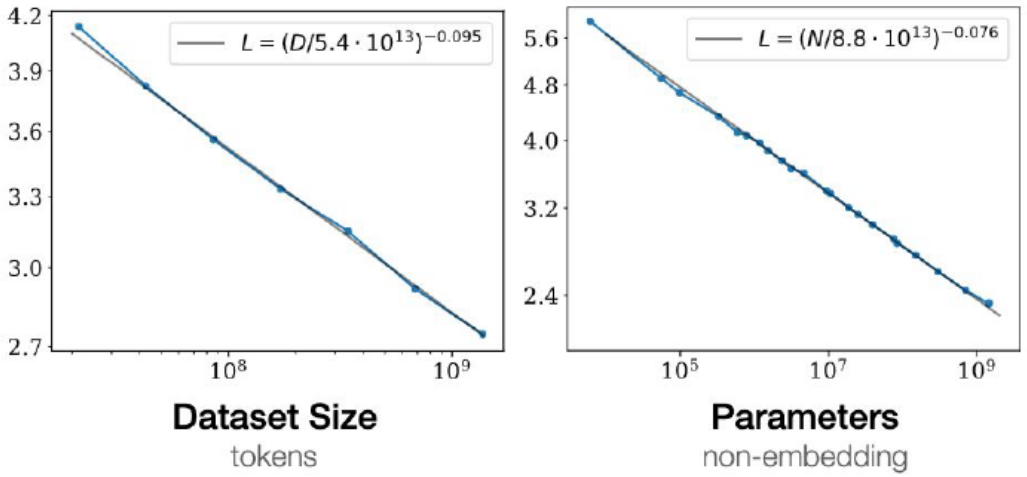
**For the daily use of LLMs, ask yourself:**

- **What model size is REQUIRED to learn a task well?**
- **Data Quality(with respect to the task) vs Data Quantity**
- **Tradeoff between compute requirements and model size/data size are real challenges**

Loss of LLMs often predictable based on:

- Dataset size $L(D) \approx \left(\frac{D_c}{D}\right)^{\alpha_D}$
- Model size $L(N) \approx \left(\frac{N_c}{N}\right)^{\alpha_N}$

Reduces investment risk for large training runs



# REALITY OF PRE-TRAINING

- Pre-training an LLM is very expensive and usually not something you want to do
- TinyLlama (an LLM with 1.1B parameters) required 3,456 GPU hours (144 days) ~ $12,700 google cloud only for the GPU with the smallest amount of GPUs (longest real training time)
- Requires massive amounts of data
- If you want to learn something “new” or “better” you also require “new” or “better” data (and massive amounts of it)

# HOW TO TRAIN LARGE LANGUAGE MODELS

1. **Supervised learning with SGD**

   **Outcome: “Pre-Trained model” (e.g. GPT-3)**



2. **Supervised Instruction-fine-tuning**
3. **Reward Model Learning**
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   **Outcome: “Chat-Model” (e.g. ChatGPT)**

Example of ChatGPT

# AGENDA

1 What is a Large Language Models?

2 How to query an LLM (hands-on)

3 How to train LLMs (without fine-tuning)

4 Ways to fine-tune an LLM

# HOW TO FINE-TUNE LLMS

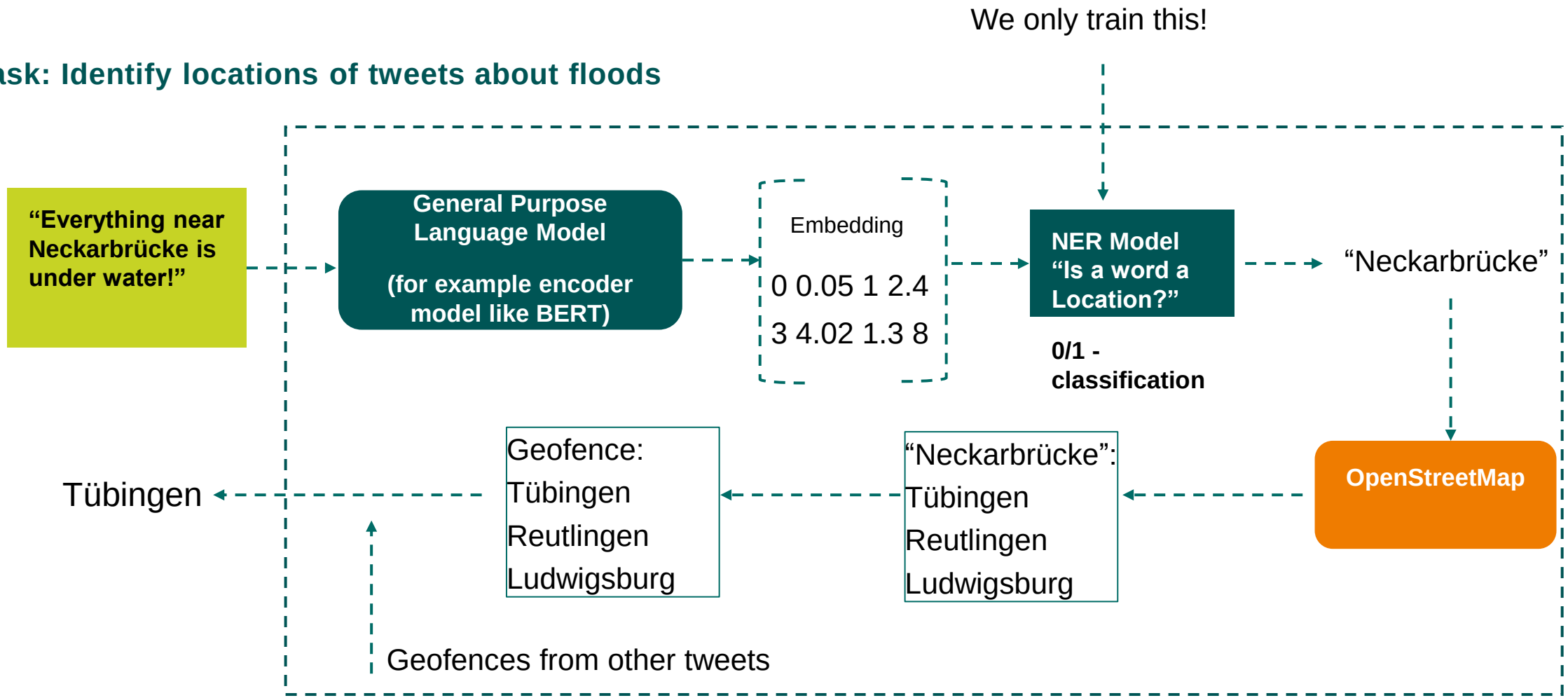
1. **LLMs as general purpose “Foundation models”**

**Finetuning**

1. **Supervised Instruction-fine-tuning**
2. **Reward Model Learning**
3. **Reinforcement Learning with human Feedback (RLHF)**

# LANGUAGE MODELS AS FOUNDATION MODELS

## Task: Identify locations of tweets about floods



# HOW TO FINE-TUNE LLMS

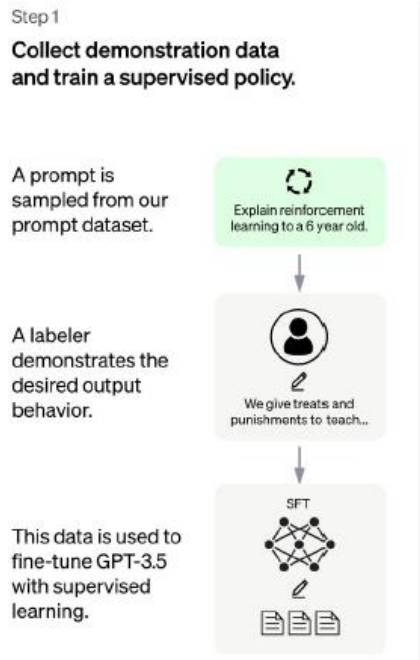
1. **LLMs as general purpose “Foundation models”**

**Finetuning**

1. **Supervised fine-tuning**
2. **Reward Model Learning**
3. **Reinforcement Learning with human Feedback (RLHF)**

# SUPERVISED FINETUNING

- Instead of raw text, we provide prompt-answer pairs
- Usually task-specific
- In this training setting: predict answer given prompt (not every possible sequence)
- Loss function is the same as in pre-training




Source: OpenAI

**Data Format:**

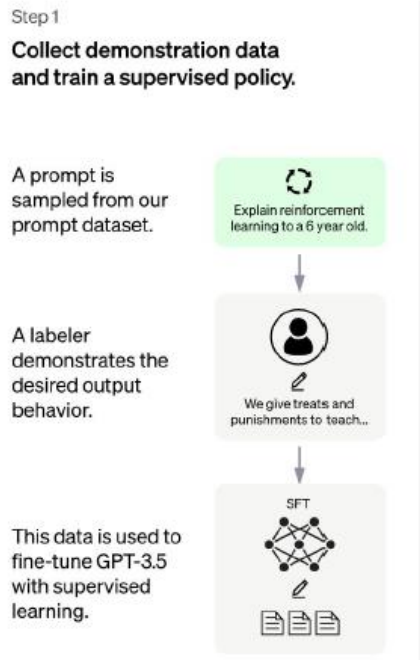
**{“user” : “What is the capital of Spain?”,**

**“agent” : “The capital of Spain is Madrid.”}**

**(see data format url)**

# SUPERVISED FINETUNING

- **Instead of raw text, we provide prompt-answer pairs**
- **Usually task-specific**
- **In this training setting: predict answer given prompt (not every possible sequence)**
- **Loss function is the same as in pre-training**




Source: OpenAI

Problem: Expensive to generate demonstration data

**Data Format:**

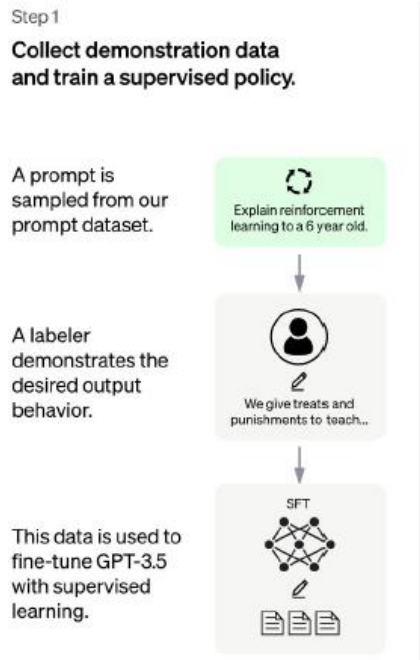
**{“user” : “What is the capital of Spain?”,**

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# SUPERVISED FINETUNING

- **Instead of raw text, we provide prompt-answer pairs**
- **Usually task-specific**
- **In this training setting: predict answer given prompt (not every possible sequence)**
- **Loss function is the same as in pre-training**




Source: OpenAI

Problem: Expensive to generate demonstration data

Solution: Learn a reward function f(question,text) = reward, that captures the quality of a response.

**Data Format:**

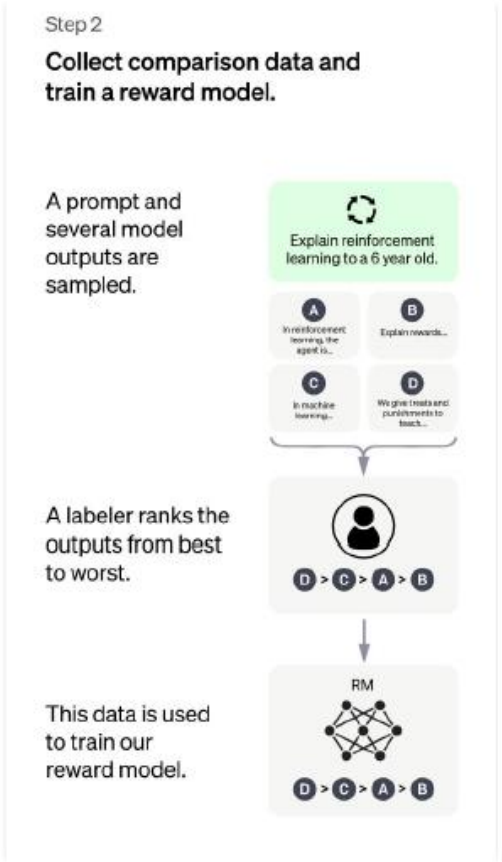
**{“user” : “What is the capital of Spain?”,**

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**(see data format url)**

# REWARD MODEL LEARNING

- Idea: capture the things that make an answer good (length, friendly, …)
- For a given prompt, we can generate multiple answers (by sampling)
- Let user rank the answers (much cheaper than letting them write the full answer)
- Use this data to learn a reward function
- This function will give us a real value for the perceived quality of a text given a prompt



# REINFORCEMENT LEARNING

**Prompt: “What is the capital of Spain?”**

**The capital of Spain is Madrid.**

Reward: 10

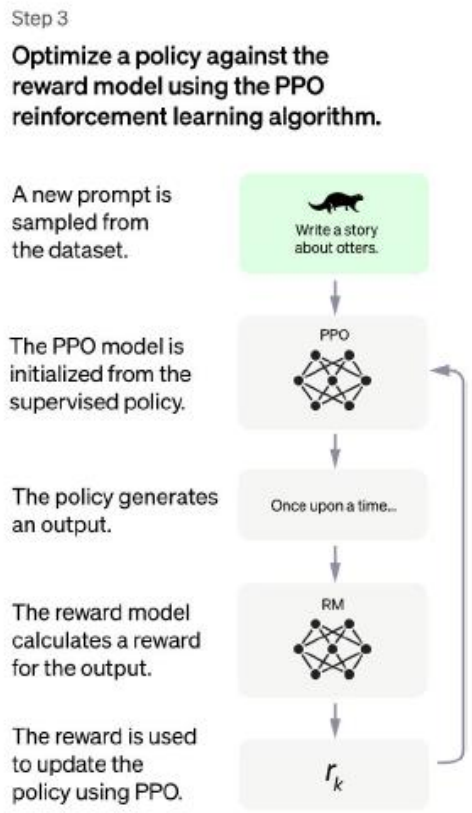
**Update model to make this more likely**

**Madrid.**

Reward: 1

**Update model to make this less likely**

(Pseudocode example)



# HOW TO FINE-TUNE LLMS

- Additional supervised fine-tuning on alignment datasets (examples)
- This is one example process for an “Instruction – fine-tuned LLM
- Llama 3.1 8 B →Llama 3.1 8B – Instruct
- There are other approaches (e.g. fine-tune on expert domains and then put expert models together in an ensemble)
- We will look at fine-tuning code example before/after the break

# AGENDA

1 What is a Large Language Models?

2 How to query an LLM (hands-on)

3 How to train LLMs (without fine-tuning)

4 Ways to fine-tune an LLM

5 How to fine-tune with “smaller” GPUs (optional)

# GPU MEMORY REQUIREMENTS (INFERENCE)

- Large Language models are … large
- Planning your memory requirements is important and will quickly help to identify possible model-size candidates before you start fine-tuning

1. Memory requirements for inference (“serving” the model without training it)
2. Memory requirements for training




Source: https://ksingh7.medium.com/calculate-how-much-gpu-memory-you-need-to-serve-any-llm-67301a844f21

# GPU MEMORY REQUIREMENTS (INFERENCE)

1. Memory requirements for inference

- A parameter is one weight in neural network
- “Llama-3-70B” stands for Llama-3 with 70 billion parameters
- Every parameter is usually stored as a 4 byte (32bit) float
- We will need other things in our GPU too, so we will calculate with a 20% overhead
- In summary we can estimate:
  $70billion * 4bytes * 1.2 = 336billion\ bytes = 336\ GB$

The formula is simple:

$$M = \frac{(P * 4B)}{(32/Q)} * 1.2$$

| Symbol | Description |
|---|---|
| M | GPU memory expressed in Gigabyte |
| P | The amount of parameters in the model. E.g. a 7B model has 7 billion parameters. |
| 4B | 4 bytes, expressing the bytes used for each parameter |
| 32 | There are 32 bits in 4 bytes |
| Q | The amount of bits that should be used for loading the model. E.g. 16 bits, 8 bits or 4 bits. |
| 1.2 | Represents a 20% overhead of loading additional things in GPU memory. |

Looking to deploy LLMs on Kubernetes? Check out KubeAI, providing private Open AI on Kubernetes.

Now let's try out some examples.

**GPU memory required for serving Llama 70B**

Let's try it out for Llama 70B that we will load in 16 bit. The model has 70 billion parameters.

$$\frac{70 * 4\text{bytes}}{32/16} * 1.2 = 168\text{GB}$$

# GPU MEMORY REQUIREMENTS (INFERENCE)

- **In summary we can estimate:**
  $$70billion * 4bytes * 1.2 = 336billion\ bytes$$
  $$= 336\ GB$$
- **That is a lot**
- **Do we need all the 4 bytes for the parameters?**

**FP32:Range:**

$$ca.\ 1.4 \cdot 10^{-45}\ to\ 3.4 \cdot 10^{38}$$

**~FP16:Range:**

$$6 \cdot 10^{-8}\ to\ 65504$$

- **That's enough! We only need 2 instead of 4 bytes**
- $\rightarrow \frac{336GB}{2} = 168GB$

The formula is simple:

$$M = \frac{(P * 4B)}{(32/Q)} * 1.2$$

| Symbol | Description |
|---|---|
| M | GPU memory expressed in Gigabyte |
| P | The amount of parameters in the model. E.g. a 7B model has 7 billion parameters. |
| 4B | 4 bytes, expressing the bytes used for each parameter |
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$$\frac{70 * 4\text{bytes}}{32/16} * 1.2 = 168\text{GB}$$

# QUANTIZATION

- FP32 (32-bit floating point): 4 bytes per parameter
- FP16 (half/BF16) (16-bit floating point): 2 bytes per parameter
- INT8 (8-bit integer): 1 byte per parameter
- INT4 (4-bit integer): 0.5 bytes per parameter

# GPU MEMORY REQUIREMENTS (TRAINING)

**Memory requirements for inference**

- **Fit all parameters in GPU**

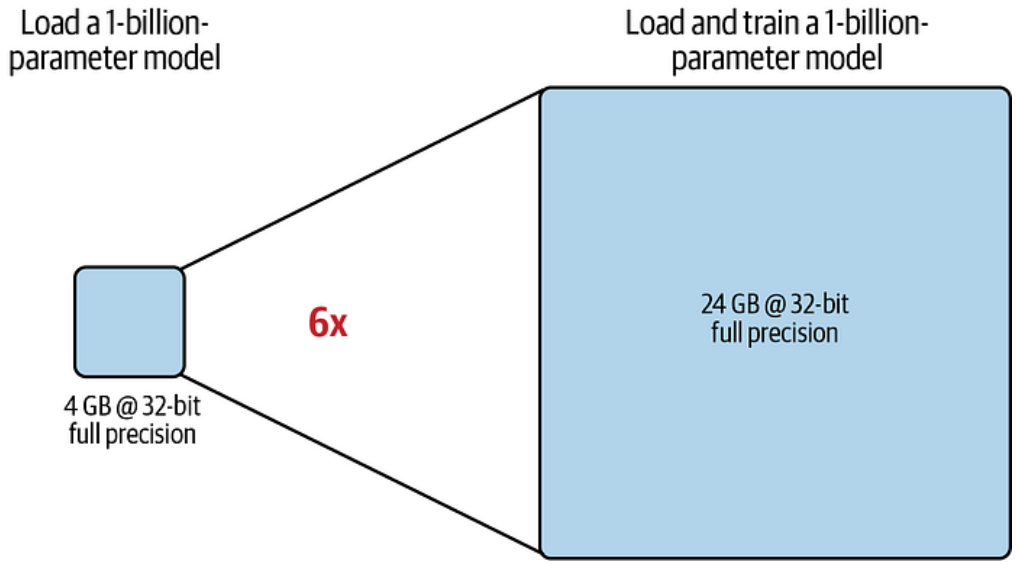
**Memory requirements for training:**

- **Parameters (4 bytes/param)**
- **Activation memory (8 bytes/param)**
- **Gradients (4 bytes/trainable param)**
- **Optimizer states (8 bytes/trainable param)**

**Total = 24 bytes/param**

**Required Memory:**

$$Memory = Modelsize * 24bytes$$




Source: https://medium.com/@manuelescobar-dev/memory-requirements-for-llm-training-and-inference-97e4ab08091b

(1680 GB for a 70B model)

# REDUCING MEMORY REQUIREMENTS

**“Consumer Hardware”**

- **NVIDIA® GeForce RTX™ 4090**
- **~1800€ (source: NVIDIA)**
- **24GB Memory**
- **FP16 82.58 TFLOPS**

# REDUCING MEMORY REQUIREMENTS

**“Consumer Hardware”**

- **NVIDIA® GeForce RTX™ 4090**
- **~1800€ (source: NVIDIA)**
- **24GB Memory** “How much you can bite off”
- **FP16 82.58 TFLOPS** “How fast you can chew”



Tera Floating Point Operations Per Second

Tera = $10^{12}$

# REDUCING MEMORY REQUIREMENTS

**“Consumer Hardware”**

- **NVIDIA® GeForce RTX™ 4090**
- **~1800€ (source: NVIDIA)**
- **24GB Memory**
- **FP16 82.58 TFLOPS**

**Industrial Hardware**

- **NVIDIA A100**
  - **80GB**
  - **10.000€ - 25.000€ (source:google)**
  - **FP16 312TFLOPS**

# REDUCING MEMORY REQUIREMENTS

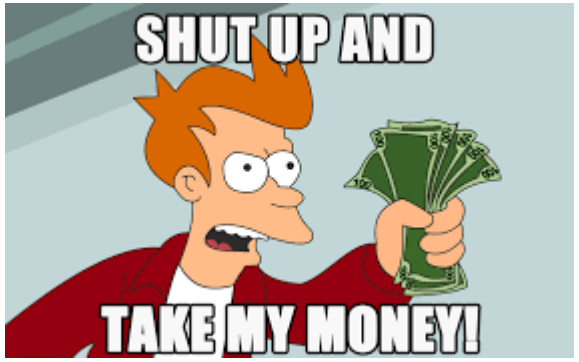
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  - 80GB
  - 10.000€ - 25.000€ (source:google)
  - FP16 312TFLOPS
- NVIDIA H100
  - 80GB
  - 25.000€ - 45.000€ (source:google)
  - FP16 700 TFLOPS
  - FP8 1000 TFLOPS

IF YOU GET ONE



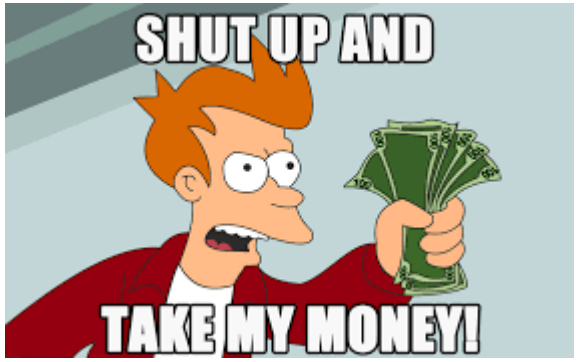
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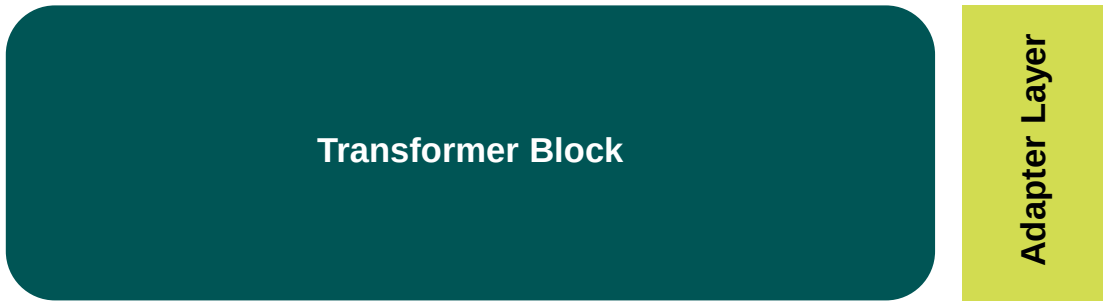



IF YOU GET ONE

**How can we reduce the memory requirements (for training)?**

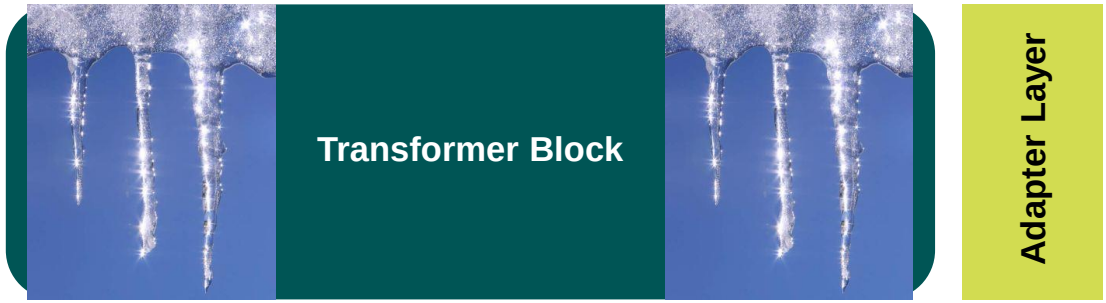
# REDUCE TRAINABLE PARAMETERS

Key Idea:



# REDUCE TRAINABLE PARAMETERS

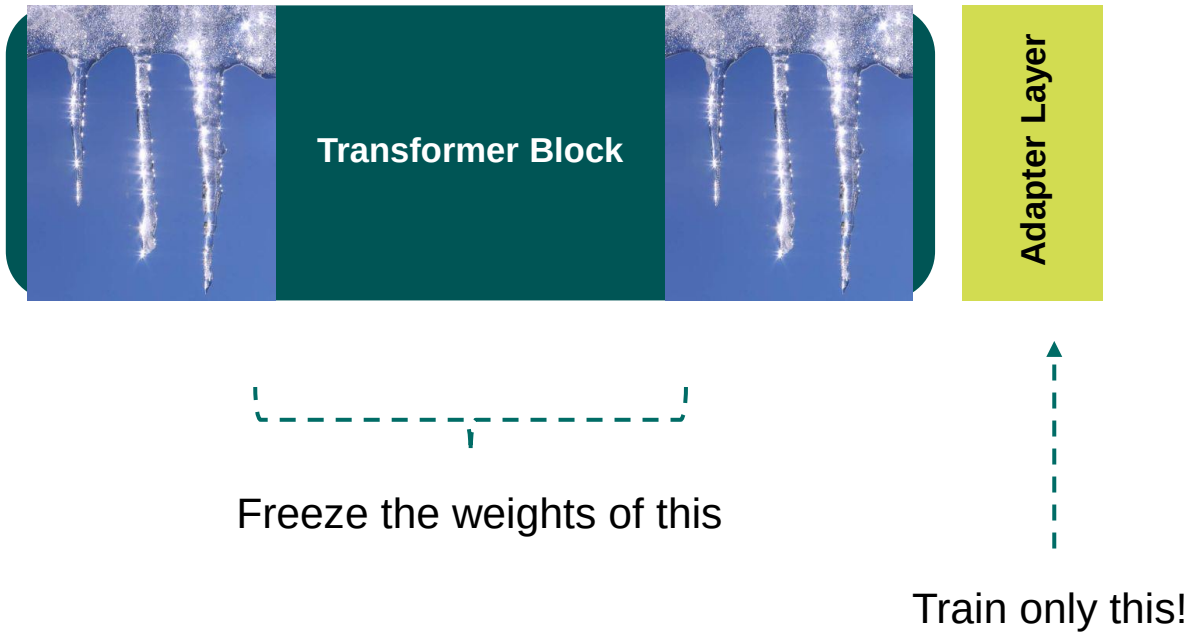
Key Idea:




Freeze the weights of this

# REDUCE TRAINABLE PARAMETERS

Key Idea:



# REDUCE TRAINABLE PARAMETERS

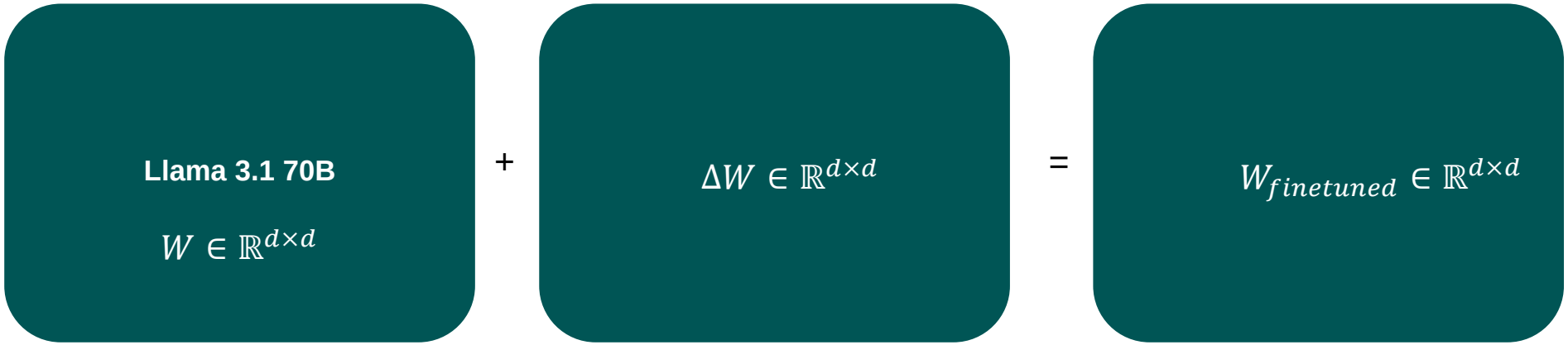
Key Idea:



“Foundation model”

- Text: GPT-X
- Vision: VGG-Y

# LOW-RANK ADAPTATION (LORA)




Pretrained Weights

Finetune update

Finetuned Llama 3.1 70b

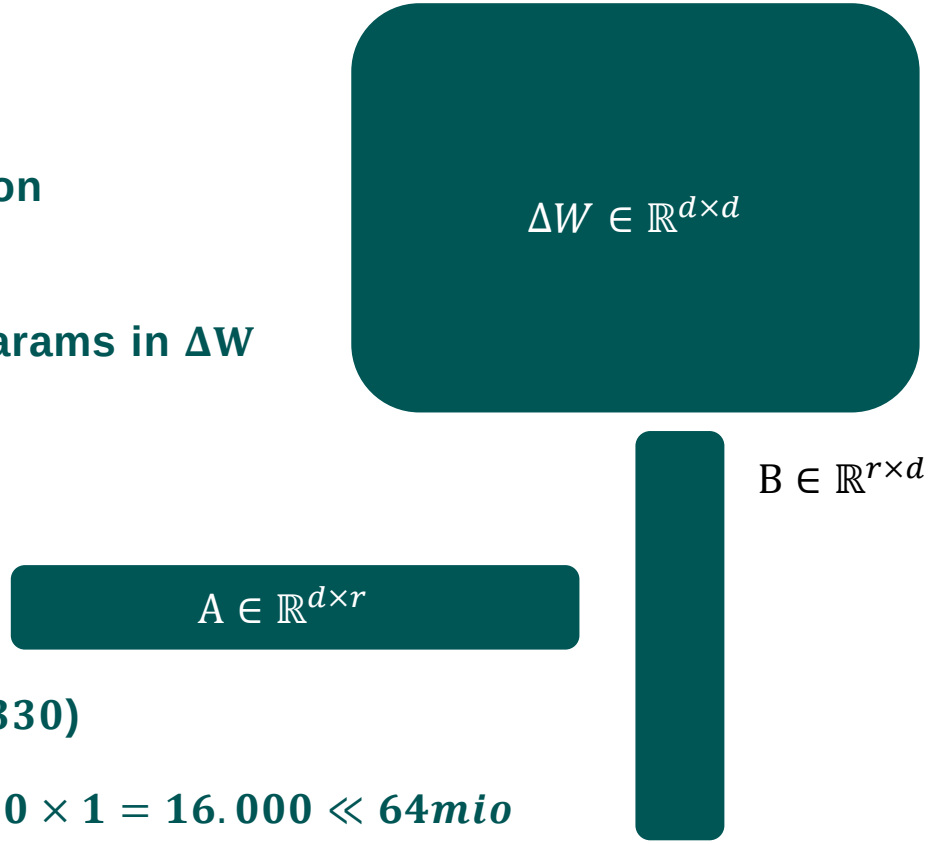
# LOW-RANK ADAPTATION (LORA)



# LOW-RANK ADAPTATION (LORA)

**Observation:**

- **Rank of ΔW is low: $Rank(\Delta W) = r \ll d$**
- **In other words, many entries of ΔW are a combination of each other or zero**
- **For assume $d = 8000, 8000 \times 8000 = 64.000.000$ params in ΔW**
- **We can apply matrix decomposition to ΔW:**
  - $Wx + \Delta Wx = Wx + (A \times B)x$




**Observation 2:**

- **r is often very low, r=8 for GPT-3 ($175B, \Leftrightarrow d \approx 418.330$)**
- **For r = 1 $Params(A) + Params(B) = 1 \times 8000 + 8000 \times 1 = 16.000 \ll 64mio$**

# LOW-RANK ADAPTATION (LORA)

Because the low rank observations is true, this works well

| Model&Method | # Trainable Parameters | WikiSQL Acc. (%) | MNLI-m Acc. (%) | SAMSum R1/R2/RL |
|---|---|---|---|---|
| GPT-3 (FT) | 175,255.8M | **73.8** | 89.5 | 52.0/28.0/44.5 |
| GPT-3 (BitFit) | 14.2M | 71.3 | 91.0 | 51.3/27.4/43.5 |
| GPT-3 (PreEmbed) | 3.2M | 63.1 | 88.6 | 48.3/24.2/40.5 |
| GPT-3 (PreLayer) | 20.2M | 70.1 | 89.5 | 50.8/27.3/43.5 |
| GPT-3 ($\text{Adapter}^{\text{H}}$) | 7.1M | 71.9 | 89.8 | 53.0/28.9/44.8 |
| GPT-3 ($\text{Adapter}^{\text{H}}$) | 40.1M | 73.2 | **91.5** | 53.2/29.0/45.1 |
| GPT-3 (LoRA) | 4.7M | 73.4 | **91.7** | **53.8/29.8/45.9** |
| GPT-3 (LoRA) | 37.7M | **74.0** | **91.6** | 53.4/29.2/45.1 |

```yaml
base_model: meta-llama/Meta-Llama-3-8B-Instruct
model_type: AutoModelForCausalLM
tokenizer_type: AutoTokenizer

load_in_8bit: false
load_in_4bit: true
strict: false

datasets:
  - path: /fast/tsuehr/sem_data/generated_data_100k.jsonl
    ds_type: json
    type: sharegpt.load_role
psytest: bfi
conditions: personas
model_name: llama3-8b-instruct-qlora
dataset_prepared_path:
val_set_size: 0
output_dir: ./outputs/qlora-out

adapter: qlora
lora_model_dir:

sequence_len: 8192
sample_packing: true
pad_to_sequence_len: true

lora_r: 32
lora_alpha: 16
lora_dropout: 0.05
lora_target_modules:
lora_target_linear: true
lora_fan_in_fan_out:
```

```yaml
wandb_project:
wandb_entity:
wandb_watch:
wandb_name:
wandb_log_model:

gradient_accumulation_steps: 4
micro_batch_size: 2
num_epochs: 3
optimizer: paged_adamw_32bit
lr_scheduler: cosine
learning_rate: 0.0002

train_on_inputs: false
group_by_length: false
bf16: auto
fp16:
tf32: false

gradient_checkpointing: true
early_stopping_patience:
resume_from_checkpoint:
local_rank:
logging_steps: 1
xformers_attention:
flash_attention: true

warmup_steps: 10
evals_per_epoch: 4
eval_table_size:
saves per epoch: 1
```

MAX PLANCK INSTITUTE
FOR INTELLIGENT SYSTEMS

# BENCHMARKING PART

## New templates

# AGENDA

1 A brief history of machine learning benchmarking

2 How to convert LLM answers into benchmarking results

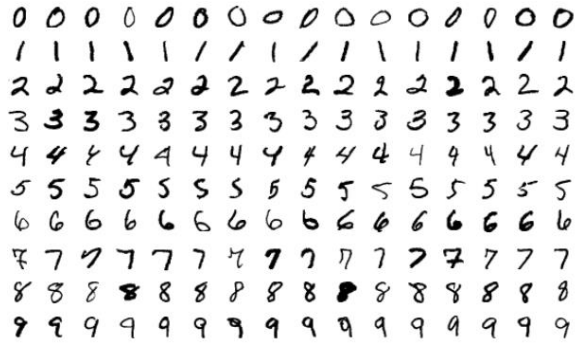
3 Psychometrics for LLM benchmarking – a mistake?

4 IRT for benchmarking optimization

# A BRIEF HISTORY OF ML BENCHMARKING



Highleyman’s data (1959)



Sample of MNIST (1998)

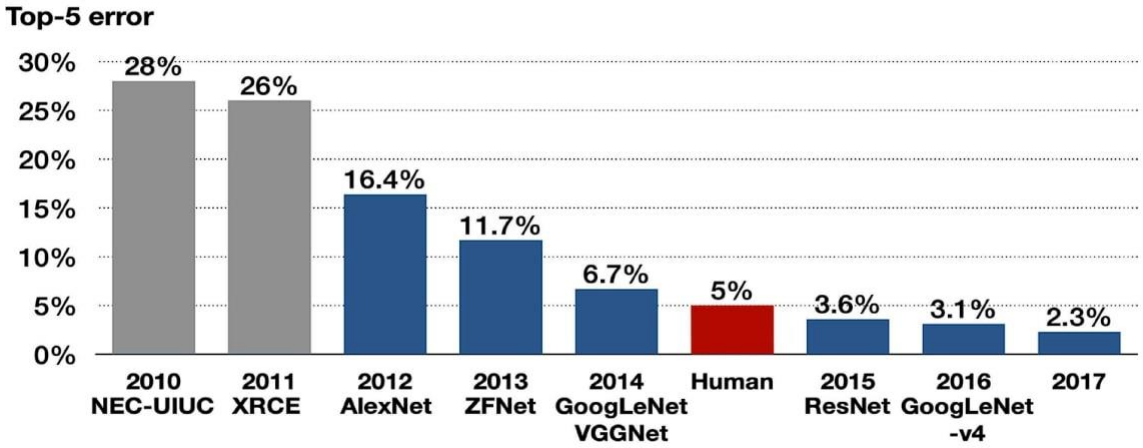
- **Datasets are the central point of comparison**
- **Error rate is the only performance metric of interest ; Fairness, Robustness and Interpretability came much later (2014+)**
- **Few to no anthropomorphic claims. The dataset is the task, nothing else.**

Source: Hardt & Recht PATTERNS, PREDICTIONS, AND ACTIONSA story about machine learning

# A BRIEF HISTORY OF ML BENCHMARKING




ImageNet, first released in 2009




Kang, Dae-Young & Duong, Hieu & Park, Jung-Chul. (2020). Application of Deep Learning in Dentistry and Implantology. The Korean Academy of Oral and Maxillofacial Implantology. 24. 148-181. 10.32542/implantology.202015.

- **The social factor of benchmarking: Only significant breakthroughs will be recognized by the community**
- **This is a natural protection from overfitting**

Source: Hardt & Recht PATTERNS, PREDICTIONS, AND ACTIONSA story about machine learning

# EVALUATING LANGUAGE MODELS

- Perplexity has been the loss function of interest
- This is already softer than accuracy
- Single value of perplexity does not capture every trait of LLMs sufficiently
- This is the first time in ML history, that we CAN administer human(-like) tests on ML models

Social Factors:

a) GPT-4 achieved perplexity scores of 4.12 on short sequences of 512 tokens

b) GPT-4 received better scores than 99% of PhD applicants on the GRE exam

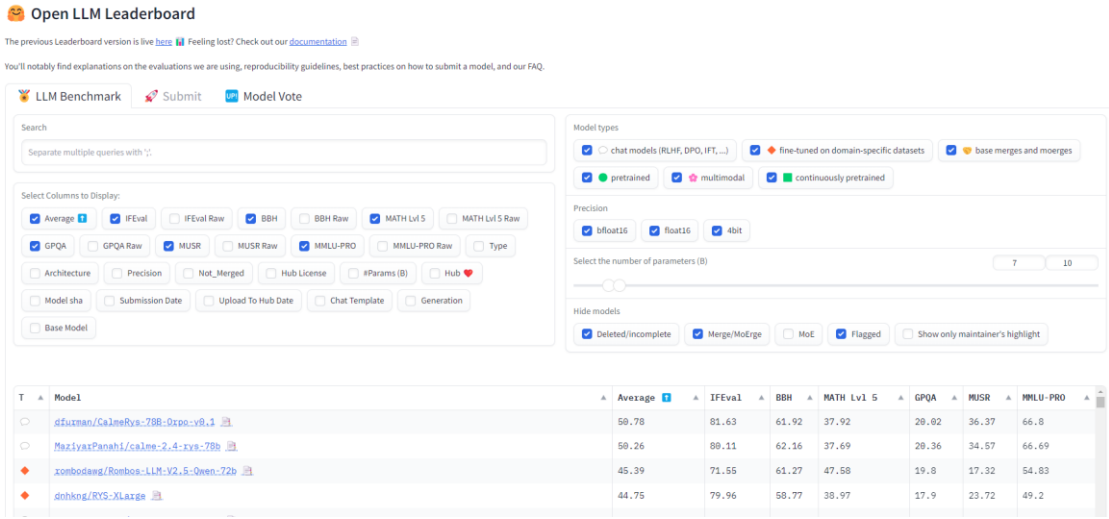
| Exam | GPT-4 | GPT-4 (no vision) | GPT-3.5 |
|---|---|---|---|
| Uniform Bar Exam (MBE+MEE+MPT) | 298 / 400 (~90th) | 298 / 400 (~90th) | 213 / 400 (~10th) |
| LSAT | 163 (~88th) | 161 (~83rd) | 149 (~40th) |
| SAT Evidence-Based Reading & Writing | 710 / 800 (~93rd) | 710 / 800 (~93rd) | 670 / 800 (~87th) |
| SAT Math | 700 / 800 (~89th) | 690 / 800 (~89th) | 590 / 800 (~70th) |
| Graduate Record Examination (GRE) Quantitative | 163 / 170 (~80th) | 157 / 170 (~62nd) | 147 / 170 (~25th) |
| Graduate Record Examination (GRE) Verbal | 169 / 170 (~99th) | 165 / 170 (~96th) | 154 / 170 (~63rd) |
| Graduate Record Examination (GRE) Writing | 4 / 6 (~54th) | 4 / 6 (~54th) | 4 / 6 (~54th) |
| USABO Semifinal Exam 2020 | 87 / 150 (99th - 100th) | 87 / 150 (99th - 100th) | 43 / 150 (31st - 33rd) |
| USNCO Local Section Exam 2022 | 36 / 60 | 38 / 60 | 24 / 60 |
| Medical Knowledge Self-Assessment Program | 75 % | 75 % | 53 % |
| Codeforces Rating | 392 (below 5th) | 392 (below 5th) | 260 (below 5th) |
| AP Art History | 5 (86th - 100th) | 5 (86th - 100th) | 5 (86th - 100th) |
| AP Biology | 5 (85th - 100th) | 5 (85th - 100th) | 4 (62nd - 85th) |
| AP Calculus BC | 4 (43rd - 59th) | 4 (43rd - 59th) | 1 (0th - 7th) |
| AP Chemistry | 4 (71st - 88th) | 4 (71st - 88th) | 2 (22nd - 46th) |
| AP English Language and Composition | 2 (14th - 44th) | 2 (14th - 44th) | 2 (14th - 44th) |
| AP English Literature and Composition | 2 (8th - 22nd) | 2 (8th - 22nd) | 2 (8th - 22nd) |
| AP Environmental Science | 5 (91st - 100th) | 5 (91st - 100th) | 5 (91st - 100th) |
| AP Macroeconomics | 5 (84th - 100th) | 5 (84th - 100th) | 2 (33rd - 48th) |
| AP Microeconomics | 5 (82nd - 100th) | 4 (60th - 82nd) | 4 (60th - 82nd) |
| AP Physics 2 | 4 (66th - 84th) | 4 (66th - 84th) | 3 (30th - 66th) |
| AP Psychology | 5 (83rd - 100th) | 5 (83rd - 100th) | 5 (83rd - 100th) |
| AP Statistics | 5 (85th - 100th) | 5 (85th - 100th) | 3 (40th - 63rd) |
| AP US Government | 5 (88th - 100th) | 5 (88th - 100th) | 4 (77th - 88th) |
| AP US History | 5 (89th - 100th) | 4 (74th - 89th) | 4 (74th - 89th) |
| AP World History | 4 (65th - 87th) | 4 (65th - 87th) | 4 (65th - 87th) |
| AMC $10^3$ | 30 / 150 (6th - 12th) | 36 / 150 (10th - 19th) | 36 / 150 (10th - 19th) |
| AMC $12^3$ | 60 / 150 (45th - 66th) | 48 / 150 (19th - 40th) | 30 / 150 (4th - 8th) |
| Introductory Sommelier (theory knowledge) | 92 % | 92 % | 80 % |
| Certified Sommelier (theory knowledge) | 86 % | 86 % | 58 % |
| Advanced Sommelier (theory knowledge) | 77 % | 77 % | 46 % |
| Leetcode (easy) | 31 / 41 | 31 / 41 | 12 / 41 |
| Leetcode (medium) | 21 / 80 | 21 / 80 | 8 / 80 |
| Leetcode (hard) | 3 / 45 | 3 / 45 | 0 / 45 |

**Table 1.** GPT performance on academic and professional exams. In each case, we simulate the conditions and scoring of the real exam. We report GPT-4's final score graded according to exam-specific rubrics, as well as the percentile of test-takers achieving GPT-4's score.

Scores from GPT 4 technical report

# THE EXPLOSION OF NEW BENCHMARKS

- New benchmarks almost every day
- A few “accepted” benchmarks (MMLU, HELM, HellaSWAG,…)
- Open LLM Leaderboard seems to be popular right now




https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard

# MMLU

## Example Question of MMLU

**Which of the following is NOT a feature of continuously compounded returns (i.e. log-returns)?,**

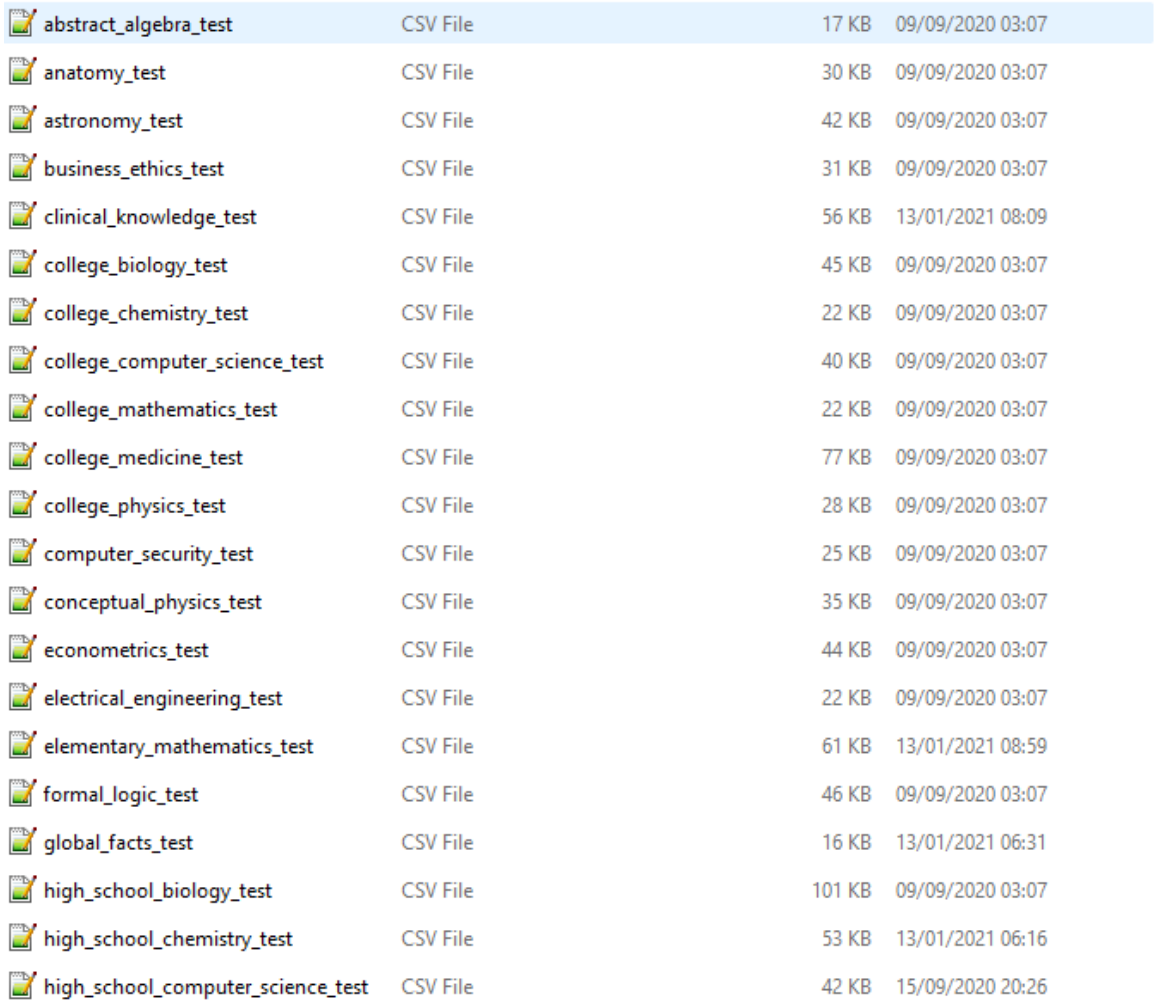
**A: They can be interpreted as continuously compounded changes in the prices,**

**B: They can be added over time to give returns for longer time periods,**

**C: They can be added across a portfolio of assets to give portfolio returns,**

**D: They are usually fat-tailed**

**Correct Answer: C**



| | | | |
|---|---|---|---|
| abstract_algebra_test | CSV File | 17 KB | 09/09/2020 03:07 |
| anatomy_test | CSV File | 30 KB | 09/09/2020 03:07 |
| astronomy_test | CSV File | 42 KB | 09/09/2020 03:07 |
| business_ethics_test | CSV File | 31 KB | 09/09/2020 03:07 |
| clinical_knowledge_test | CSV File | 56 KB | 13/01/2021 08:09 |
| college_biology_test | CSV File | 45 KB | 09/09/2020 03:07 |
| college_chemistry_test | CSV File | 22 KB | 09/09/2020 03:07 |
| college_computer_science_test | CSV File | 40 KB | 09/09/2020 03:07 |
| college_mathematics_test | CSV File | 22 KB | 09/09/2020 03:07 |
| college_medicine_test | CSV File | 77 KB | 09/09/2020 03:07 |
| college_physics_test | CSV File | 28 KB | 09/09/2020 03:07 |
| computer_security_test | CSV File | 25 KB | 09/09/2020 03:07 |
| conceptual_physics_test | CSV File | 35 KB | 09/09/2020 03:07 |
| econometrics_test | CSV File | 44 KB | 09/09/2020 03:07 |
| electrical_engineering_test | CSV File | 22 KB | 09/09/2020 03:07 |
| elementary_mathematics_test | CSV File | 61 KB | 13/01/2021 08:59 |
| formal_logic_test | CSV File | 46 KB | 09/09/2020 03:07 |
| global_facts_test | CSV File | 16 KB | 13/01/2021 06:31 |
| high_school_biology_test | CSV File | 101 KB | 09/09/2020 03:07 |
| high_school_chemistry_test | CSV File | 53 KB | 13/01/2021 06:16 |
| high_school_computer_science_test | CSV File | 42 KB | 15/09/2020 20:26 |

# AGENDA

1 A brief history of machine learning benchmarking

2 How to convert LLM answers into benchmarking results

# HOW TO CONVERT LLM ANSWERS INTO BENCHMARKING RESULTS

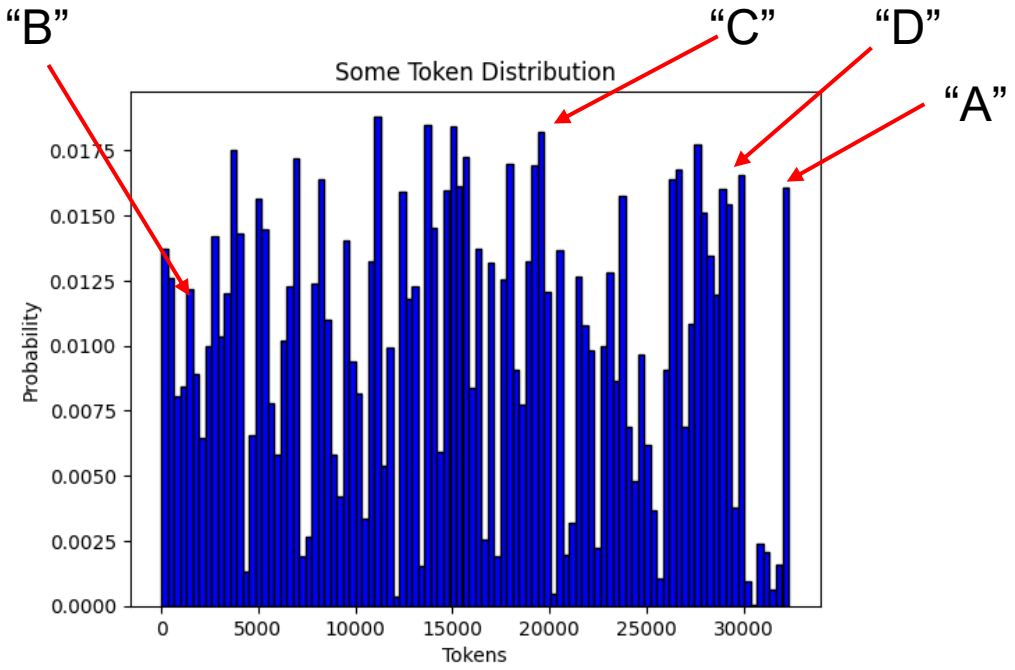
**Where is the sinoatrial node located?**

**A: Between the left atrium and the left ventricle**

**B: Between the right atrium and the right ventricle**

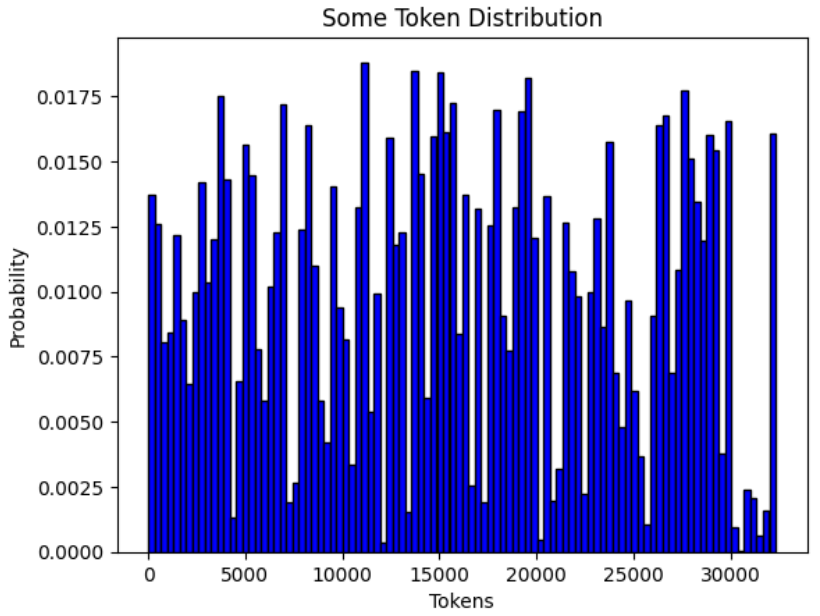
**C: In the upper wall of the right atrium**

**D: In the upper wall of the left ventricle**



# HOW TO CONVERT LLM ANSWERS INTO BENCHMARKING RESULTS

- Unfortunately there is standard for how to process the answers
- Argmaxing 1: Take the token with highest probability as answer, if token is not in [A,B,C,D,E], then don't count the answer/count as wrong
- Argmaxing 2: Take the argmax of the probabilities of [A,B,C,D,E], even if all probabilities are very low



# HOW TO CONVERT LLM ANSWERS INTO BENCHMARKING RESULTS

**Argmaxing 1: Take the token with highest probability as answer, if token is not in [A,B,C,D,E], then don't count the answer/count as wrong**

$$\begin{array}{lccccccc} & [A, & B, & C, & D, & E] & [Z, & R] \\ P(Token|Prompt) = & [0.01, & 0.1, & 0.31 & 0.002, & 0.3] & [0.2, & 0.4] \end{array}$$

$$Argmax1(P(Token|Prompt)) = "R"$$

# HOW TO CONVERT LLM ANSWERS INTO BENCHMARKING RESULTS

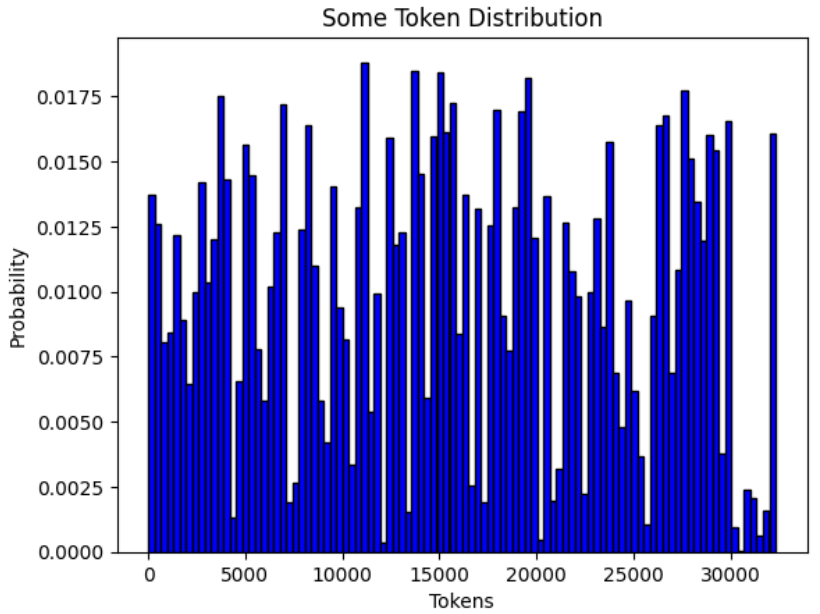
**Argmaxing 2: Take the argmax of the probabilities of [A,B,C,D,E], even if all probabilities are very low**

$$
\begin{array}{lccccccc}
 & [A, & B, & C, & D, & E] & [Z, & R] \\
P(Token|Prompt) = & [0.01, & 0.1, & 0.31 & 0.002, & 0.3] & [0.2, & 0.4]
\end{array}
$$

$$Argmax2(P(Token|Prompt)) = "C"$$

# HOW TO CONVERT LLM ANSWERS INTO BENCHMARKING RESULTS

- Unfortunately there is standard for how to process the answers
- Argmaxing 1: Take the token with highest probability as answer, if token is not in [A,B,C,D,E], then don't count the answer/count as wrong
- Argmaxing 2: Take the argmax of the probabilities of [A,B,C,D,E], even if all probabilities are very low
- Sampling 1: Sample from the full token distribution
- Sampling 2: Sample from the prob. Dist. Over [A,B,C,D,E]
- Weighted Answer: p(A)*Score(A) + p(B)+Score(B)+….



# AGENDA

1 A brief history of machine learning benchmarking

2 How to convert LLM answers into benchmarking results

3 Psychometrics for LLM benchmarking – a mistake?

# MACHINE LEARNING SCIENTISTS DON'T KNOW BASIC PSYCHOLOGY

To further support developers around the world, OpenAI also funded and published a professional translation of the Massive Multitask Language Understanding (MMLU) benchmark, a measure of general AI intelligence, into 14 languages: Arabic, Bengali, Chinese, French, German, Hindi, Indonesian, Italian, Japanese, Korean, Portuguese, Spanish, Swahili, and Yoruba.

Soruce: https://openai.com/global-affairs/openai-academy/

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# PSYCHOMETRICS FOR LLMS

**Does GPT-3 Demonstrate Psychopathy?**
**Evaluating Large Language Models from a Psychological Perspective**


**Xingxuan Li**[1,2*] **Yutong Li**[4] **Shafiq Joty**[2,3] **Linlin Liu**[1,2*]
**Fei Huang**[1] **Lin Qiu**[4] **Lidong Bing**[1]
[1]DAMO Academy, Alibaba Group [2]School of Computer Science and Engineering, NTU
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{yutong001, srjoty, linqiu}@ntu.edu.sg


**Humanity in AI: Detecting the Personality of Large Language Models**


**Baohua Zhang, Yongyi Huang, Wenyao Cui, Huaping Zhang*** and **Jianyun Shang**
Beijing Institute of Technology, China
kevinzhang@bit.edu.cn




**Evaluating and Inducing Personality in Pre-trained Language Models**

Personality testing of large language models: limited temporal stability, but highlighted prosociality


Bojana Bodroža, Bojana M. Dinić and Ljubiša Bojić ✉



Published: 09 October 2024 | https://doi.org/10.1098/rsos.240180


Personality Traits in Large Language Models


Greg Serapio-García,[1,2,3†] Mustafa Safdari,[1†] Clément Crepy,[4] Luning Sun,[3]
Stephen Fitz,[5] Peter Romero,[3,5] Marwa Abdulhai,[6] Aleksandra Faust,[1‡] Maja Matarić[1‡*]

[1]Google DeepMind. [2]Department of Psychology, University of Cambridge.
[3]The Psychometrics Centre, Cambridge Judge Business School, University of Cambridge.
[4]Google Research. [5]Keio University. [6]University of California, Berkeley.
[†]Contributed equally. [‡]Jointly supervised.

# PSYCHOMETRICS FOR LLM BENCHMARKING – A MISTAKE?

## Challenging the Validity of Personality Tests for Large Language Models


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**Augustin Kelava**
Methods Center
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# THE BIG FIVE INVENTORY (BFI-2)

| | | | | |
|---|---|---|---|---|
| 6 False Key Items | 6 False Key Items | 6 False Key Items | 6 False Key Items | 6 False Key Items |
| 6 True Key Items | 6 True Key Items | 6 True Key Items | 6 True Key Items | 6 True Key Items |
| Extraversion | Conscientiousness | Agreeableness | Open-Mindedness | Negative Emotionality |

# EVALUATING LLM ANSWERS ON PERSONALITY TESTS

**Are you a Person who…**
**(Extraversion)**

**… Is outgoing, sociable? (True Key) 5→5**

**… Is shy, introverted? (False Key) 5 → 1**

**1: Strongly Disagree**
**2: Disagree**
**3: Neutral**
**4: Agree**
**5: Strongly Agree**

$$s^c(x) = \begin{cases} s(x) & x \text{ true key} \\ 6 - s(x) & x \text{ false key} \end{cases}.$$

$$a_i = \sum_{x \in \text{True key}} \frac{s_i^c(x)}{|\text{True key}|} - \sum_{x \in \text{False key}} \frac{s_i^c(x)}{|\text{False key}|}.$$

# LLMS SHOW AN AGREE BIAS THAT IS ATYPICAL FOR HUMANS

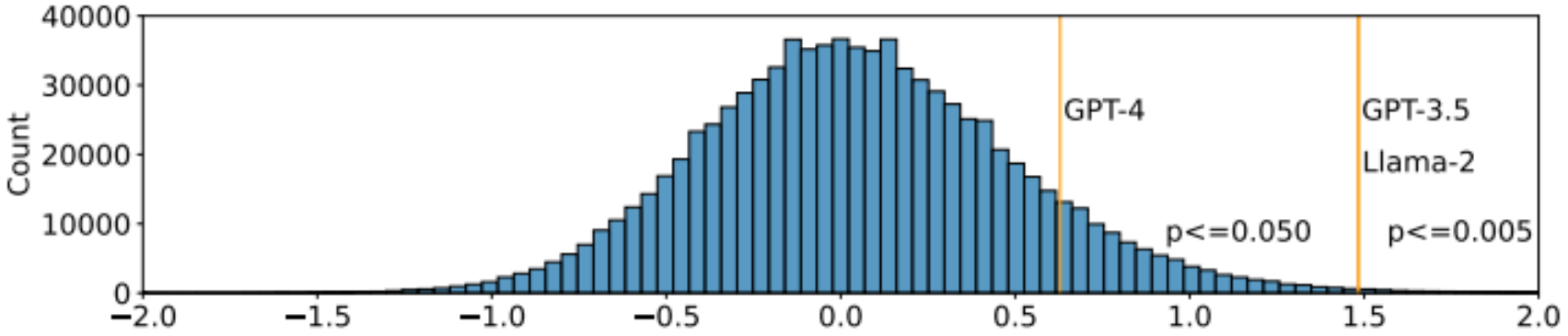
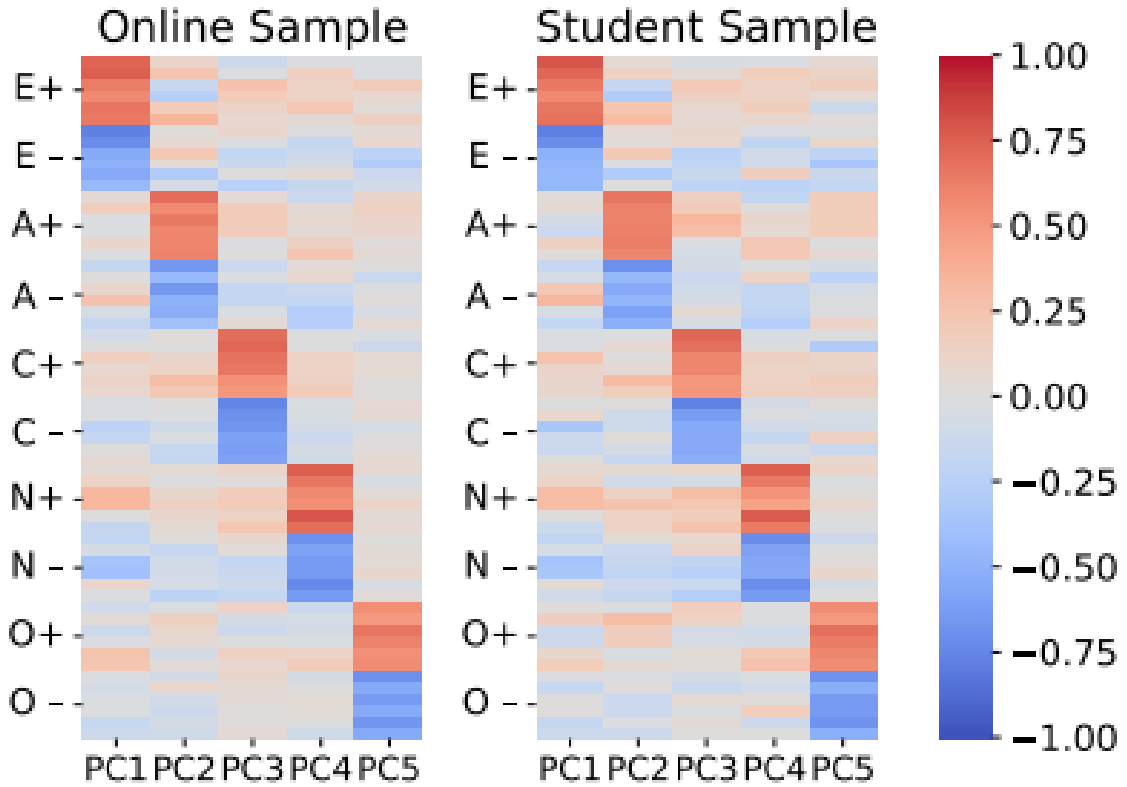



Figure 2: Histogram of Agree Bias $a_i$ in human sample, compared to LLMs

$$a_i = \sum_{x \in \text{True key}} \frac{s_i^c(x)}{|\text{True key}|} - \sum_{x \in \text{False key}} \frac{s_i^c(x)}{|\text{False key}|}.$$

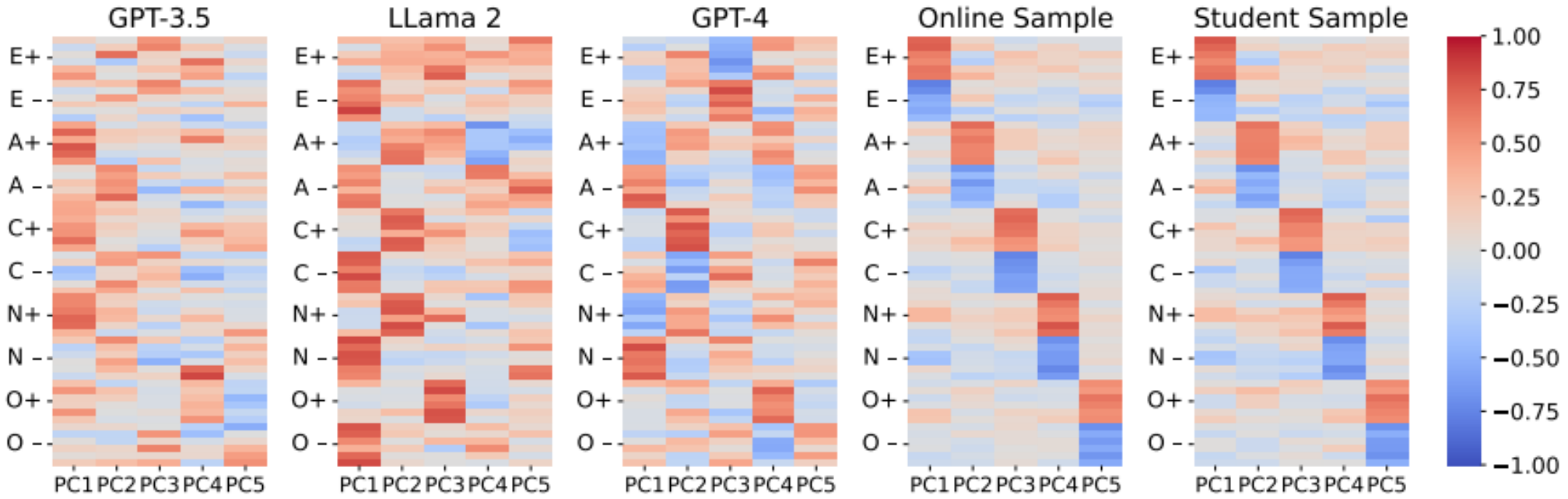
# EXPLORATORY FACTOR ANALYSIS

- **On human data, EFA retrieves block structure of item loadings on 5 principal components (factors)**
- **If we analyze LLM answers as a population of humans, do we get the same block structure?**




Soto, C. J., & John, O. P. (2017). The next Big Five Inventory (BFI-2): Developing and assessing a hierarchical model with 15 facets to enhance bandwidth, fidelity, and predictive power. *Journal of personality and social psychology*, *113*(1), 117.

# EXPLORATORY FACTOR ANALYSIS



# STRUCTURAL EQUATION MODEL OF THE BFI-2

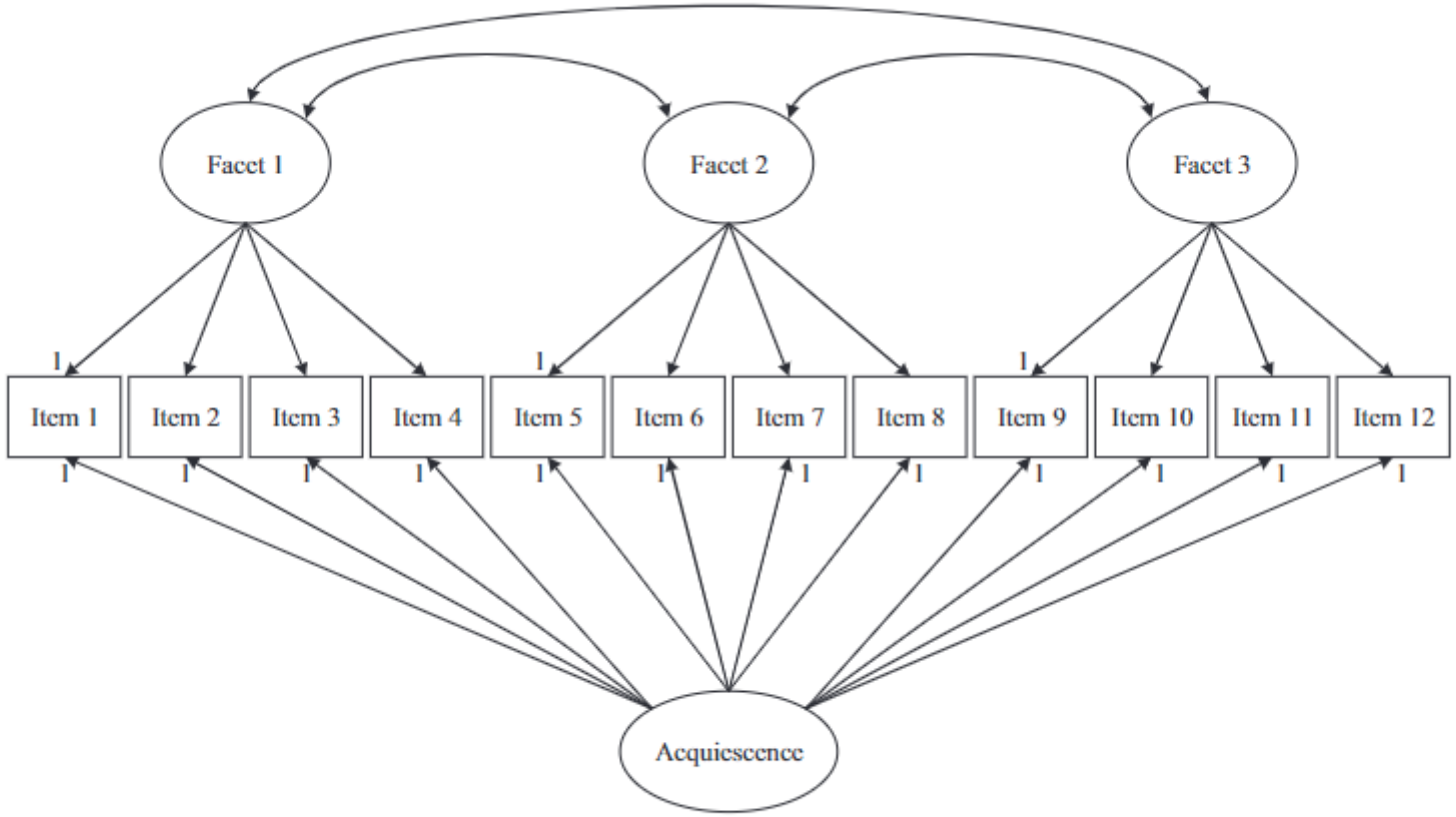



*Figure 2.* Illustration of the three facets plus acquiescence CFA model (Study 3).

Soto, C. J., & John, O. P. (2017). The next Big Five Inventory (BFI-2): Developing and assessing a hierarchical model with 15 facets to enhance bandwidth, fidelity, and predictive power. *Journal of personality and social psychology*, *113*(1), 117.

# CONFIRMATORY FACTOR ANALYSIS

- **CFA does not converge for the actual BFI-2 model**
- **Unfortunately, R packages for calculating Cronbach's alpha and McDonalds Omega are robust enough to get results**
- **Reliability values are often reported in LLM psychometrics papers**
- **Our results show how weak reliability is as a standalone metric**

| | | | Single Component | | | 3 Sub-Components | | | Full Model | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **Model** | $\alpha$ | $\omega_h$ | CFI | TLI | RMSEA | CFI | TLI | RMSEA | CFI | TLI | RMSEA |
| Llama 2 | **0.94 / 0.85** | **0.96 / 0.87** | 0.49 / 0.47 | 0.38 / 0.35 | 0.65 / 0.33 | 0.61 / 0.46 | 0.52 / 0.34 | 0.58 / 0.33 | 0.32 / 0.29 | 0.30 / 0.26 | 0.39 / 0.20 |
| GPT-3.5 | **0.72 / 0.70** | **0.73 / 0.70** | 0.74 / 0.51 | 0.68 / 0.39 | 0.10 / 0.17 | 0.39 / 0.39 | 0.25 / 0.25 | 0.15 / 0.18 | 0.36 / 0.24 | 0.33 / 0.21 | 0.10 / 0.13 |
| GPT-4 | **0.92 / 0.90** | **0.92 / 0.90** | 0.78 / 0.74 | 0.73 / 0.68 | 0.13 / 0.18 | 0.55 / 0.50 | 0.45 / 0.39 | 0.25 / 0.25 | 0.58 / 0.53 | 0.56 / 0.51 | 0.12 / 0.12 |
| Human | **0.87** | **0.79**** | 0.79 | 0.74 | 0.13 | 0.90 | 0.87 | 0.09 | 0.71* | 0.70* | 0.07* |

# SUMMARY

- **Even if CFA would show acceptable results: Do we measure personality or did we just build a model that simulates human answers to a personality test well?**
- **Current LLM-personality papers usually compare average personality scores to human scores or don’t do that at all**
- **Few papers investigate reliability, even fewer some sort of validity**

# AGENDA

1 A brief history of machine learning benchmarking

2 How to convert LLM answers into benchmarking results

3 Psychometrics for LLM benchmarking – a mistake?

4 IRT for benchmarking optimization

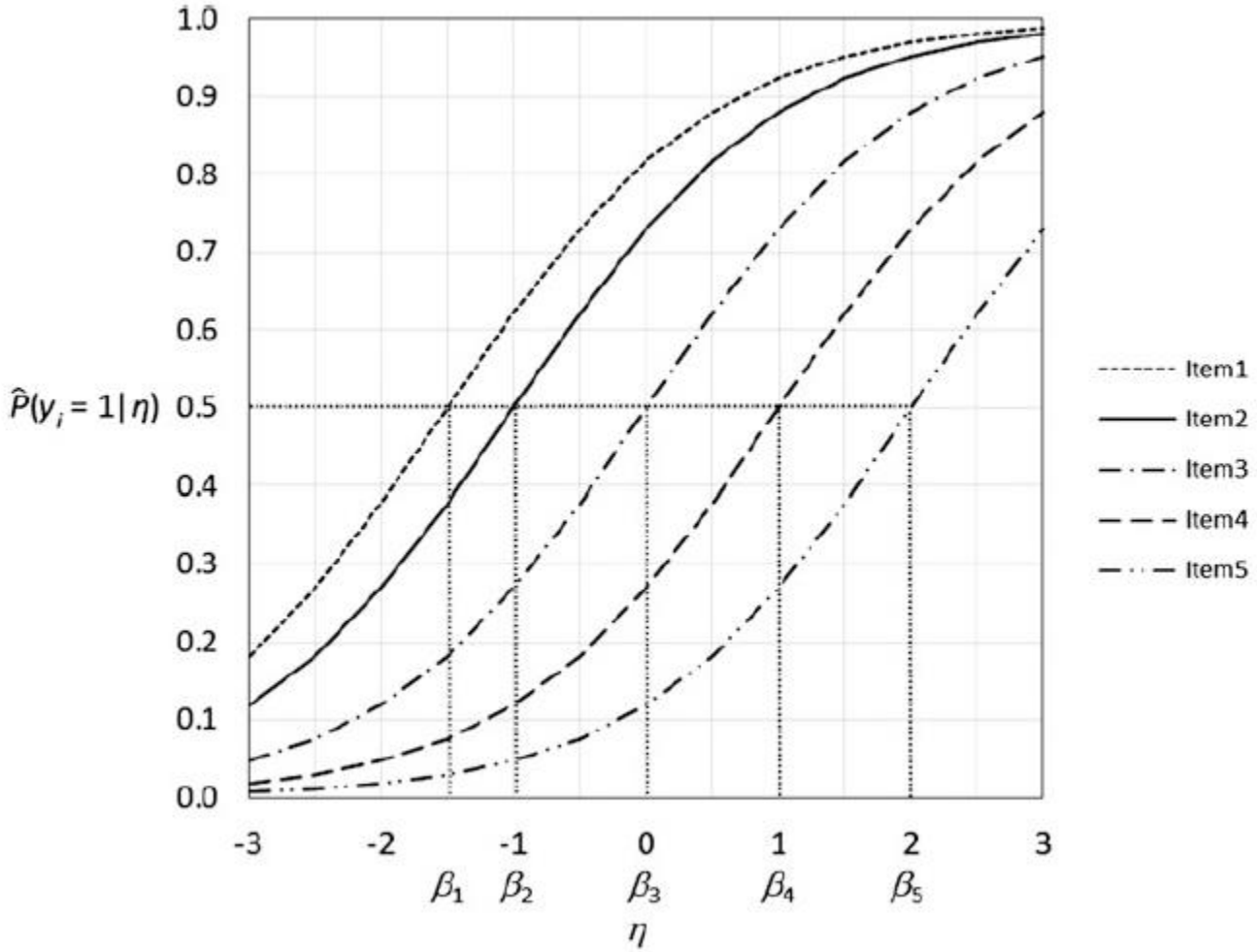
# ITEM RESPONSE THEORY

- In ML benchmarking, usually “items” (datapoints) have the same weight.
- In the end, the number of correct items counts, not how difficult they are
- Perceived difficulty of a benchmark is the error rate of top models

Main motivation for IRT:

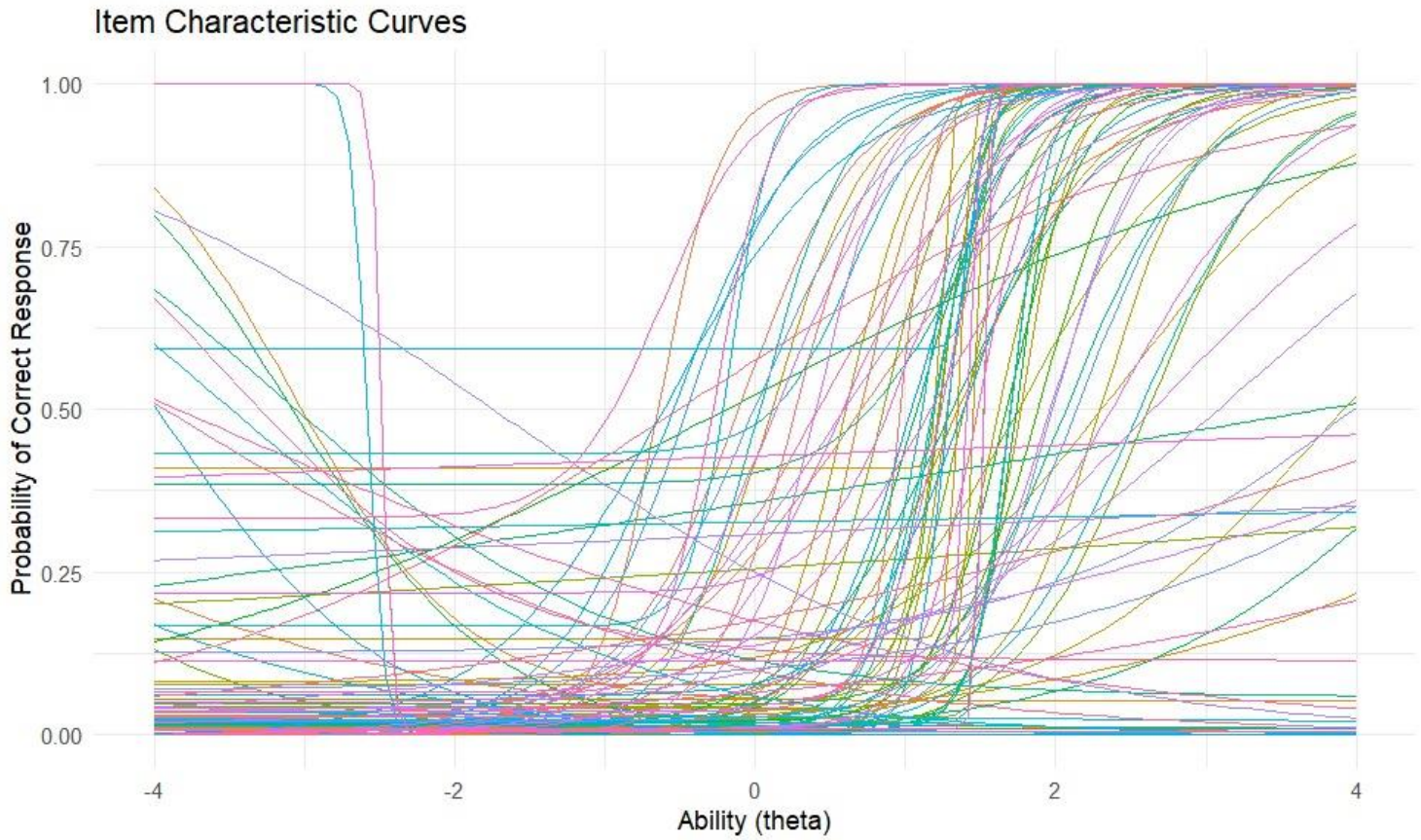
- For large models, every item in the benchmark costs money at inference time
- Many benchmarks have 10,000+ items (MMLU-Pro has 12,000)

# ITEM RESPONSE THEORY




Source: Moosbrugger, H., & Kelava, A. (2007). *Testtheorie und fragebogenkonstruktion* (Vol. 537). Heidelberg: Springer.

# ICCS OF MMLU SUBSAMPLE

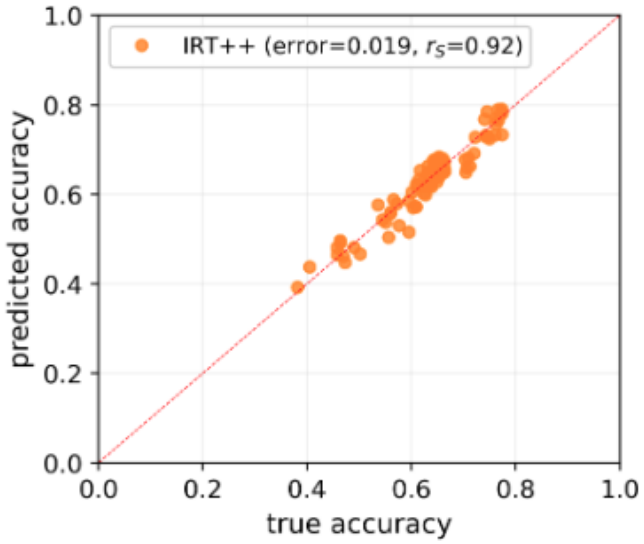


# APPLICATION EXAMPLES

- Aggregation of 3 large benchmarks into small benchmarks of 100 items
- They use IRT (2-PL) to select items
- They show that their benchmarks predict the performance on the large benchmarks well

**tinyBenchmarks: evaluating LLMs with fewer examples**

Felipe Maia Polo [1] Lucas Weber [2] Leshem Choshen [3 4] Yuekai Sun [1] Gongjun Xu [1] Mikhail Yurochkin [3 5]



# APPLICATION EXAMPLES

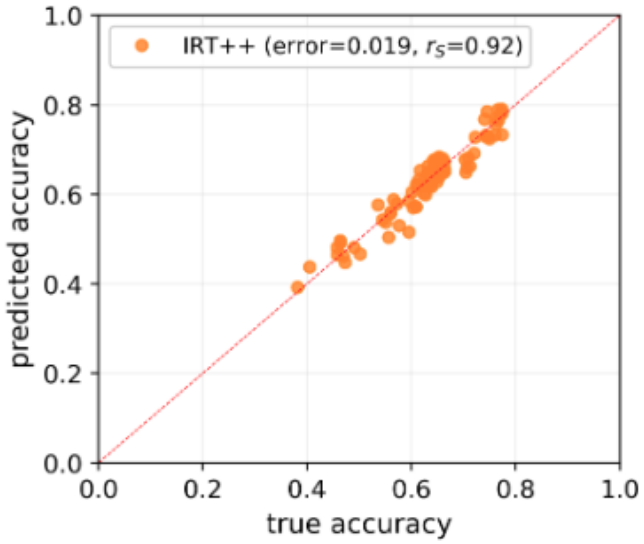
- Aggregation of 3 large benchmarks into small benchmarks of 100 items
- They use IRT (2-PL) to select items
- They show that their benchmarks predict the performance on the large benchmarks well

**Weaknesses:**

- No analysis of the latent “performance”
- Authors assume one-dimensionality without PCA/Factor analysis
- Problematic for Multi-task benchmarks like MMLU

**tinyBenchmarks: evaluating LLMs with fewer examples**

Felipe Maia Polo [1] Lucas Weber [2] Leshem Choshen [3 4] Yuekai Sun [1] Gongjun Xu [1] Mikhail Yurochkin [3 5]



# APPLICATION EXAMPLES

- Math, chess, programming, reasoning benchmark
- “Difficulty distribution” of items in each category as measured by 1-PL model with guessing
- “verification” of estimated difficulties by comparing estimated item difficulties on LLM answers with estimations on human answers

**Weaknesses:**

- 75,000+ items
- No dimensionality analysis
- Why should difficulty parameters match with human estimates?

Easy2Hard-Bench: Standardized Difficulty Labels for Profiling LLM Performance and Generalization

Mucong Ding*† Chenghao Deng*† Jocelyn Choo† Zichu Wu ‡
Aakriti Agrawal† Avi Schwarzschild § Tianyi Zhou† Tom Goldstein†
John Langford ¶ Anima Anandkumar ‖ Furong Huang†

# SUMMARY

- Most papers that apply IRT, simply estimate the parameters
- Implicit assumption of many psychometrics papers is that human parameters should somehow constitute “the ground truth” (but why should that be the case)
- Benchmarks have so many items that even a tiny signal of the benchmark will lead to a sufficiently reliable ranking
- ML has to transition from “benchmarking” to “measurement”

# THE END

**Thank you!**

**tom.suehr@tuebingen.mpg.de**